

# A 94Hz Inference and 7.4mJ/epoch Fine-tune Edge SoC for Diffusion- based Robot Manipulation with Speculation and Disturbance Enhancement

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Huazhong Yang<sup>1,2</sup>, Yongpan Liu<sup>1,2</sup>, and Hongyang Jia<sup>1,2</sup>



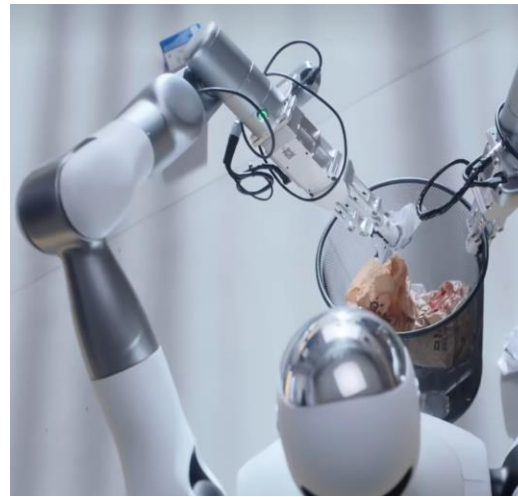
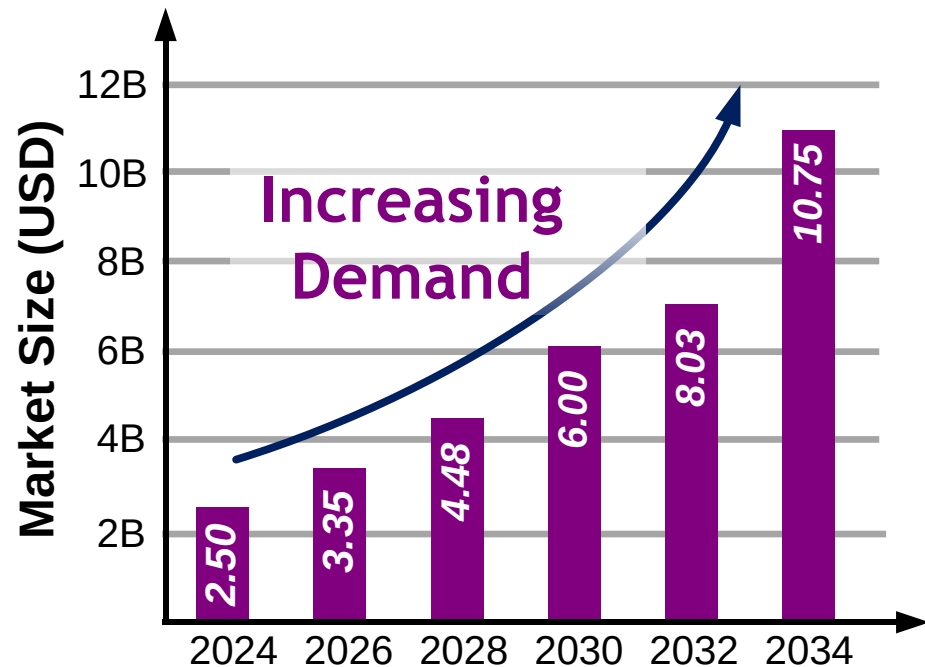
<sup>1</sup> Department of Electronic Engineering

<sup>2</sup> BNRist

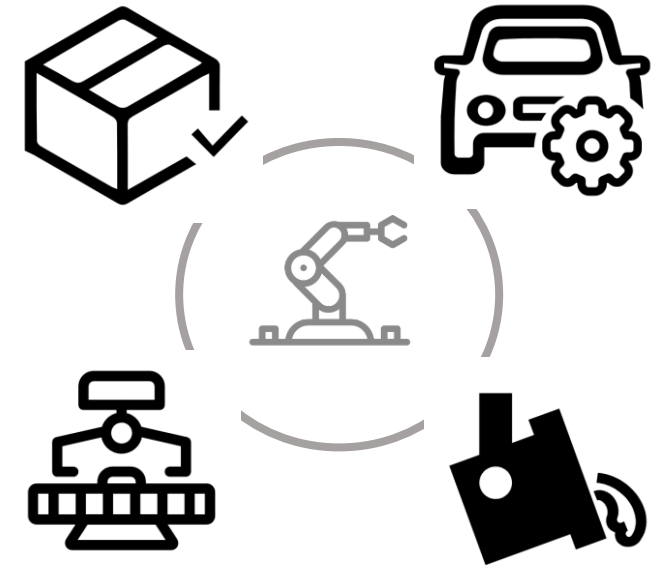
Tsinghua University, Beijing, China

# Embodied AI

- Being able to think and interact with the physical world
  - Enable a wide range of real-world applications



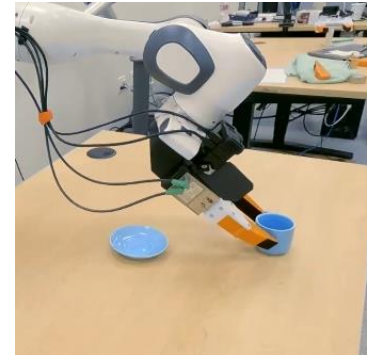
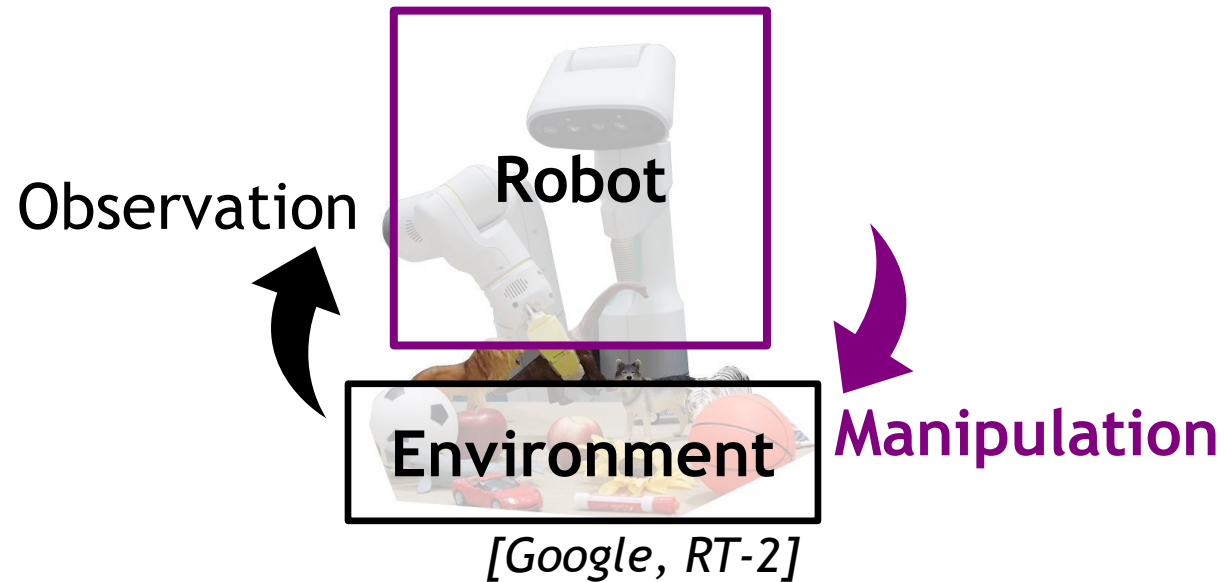
Smart Home



Manufacturing

# Robot Manipulation

- Manipulation is one of the most critical abilities of embodied AI
  - Interact with environment
  - Execute complex tasks



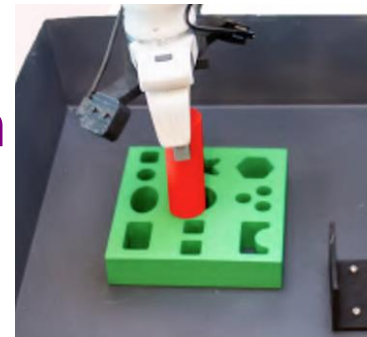
[Chi,  
Stanford,  
RSS'24]

Grasping



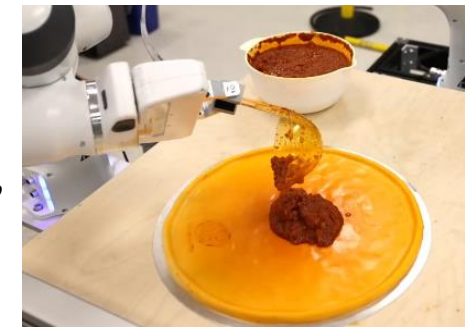
[Nvidia,  
GR00T N1]

Packing



[Zhang,  
Berkeley,  
ICRA'22]

Insertion



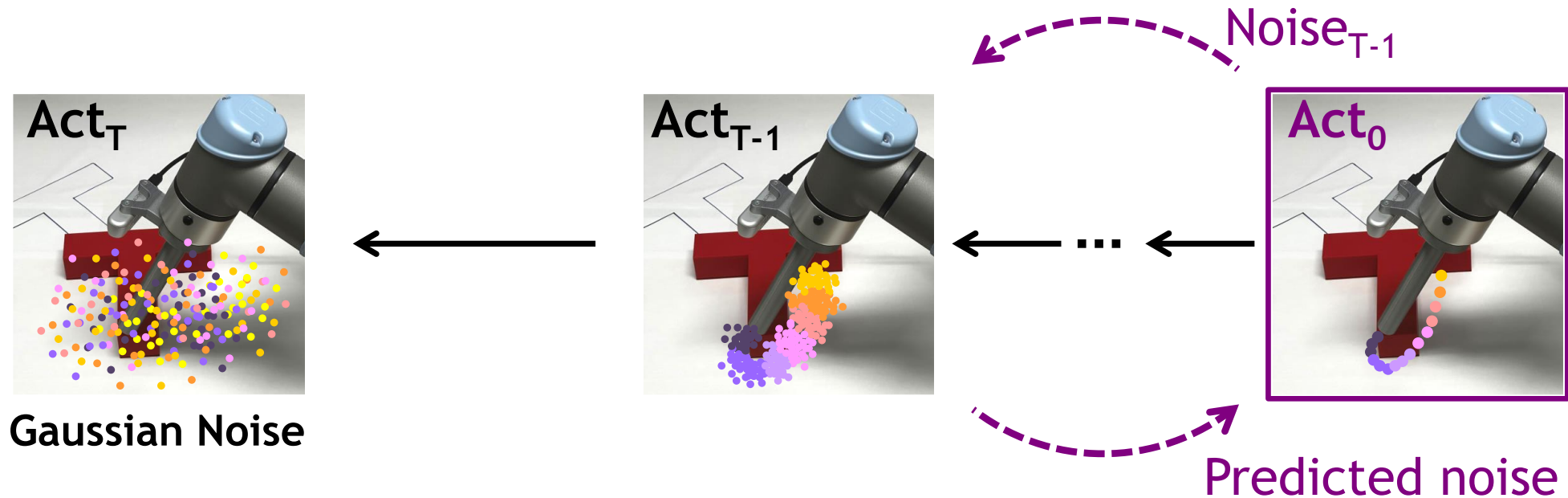
[Chi,  
Stanford,  
RSS'23]

Pouring

# Diffusion-based Action Generation

- Emerging methodology for robotic action generation (AG)
  - Better temporal action consistency, better training stability...

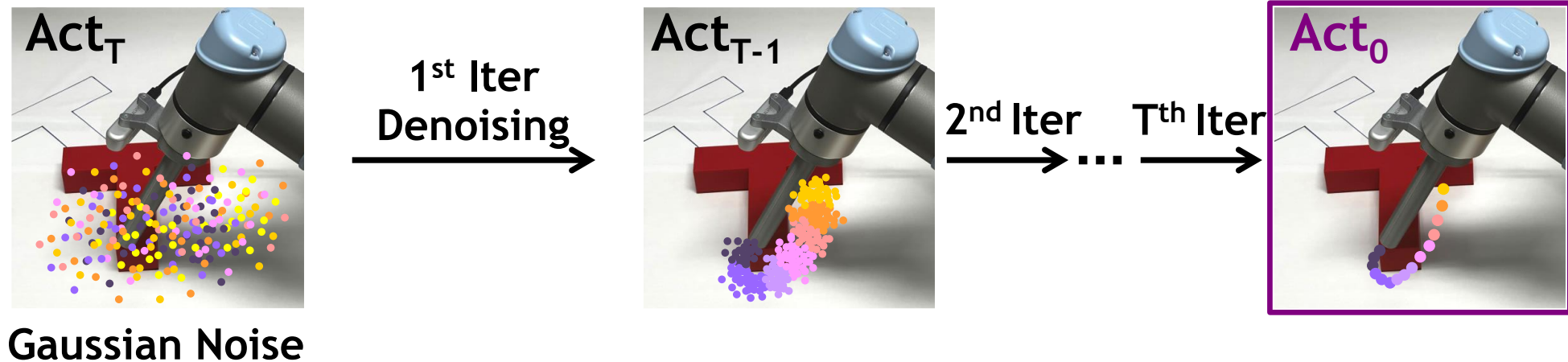
Training: Multi-step noising mathematically w/ 1 computing iteration



# Diffusion-based Action Generation

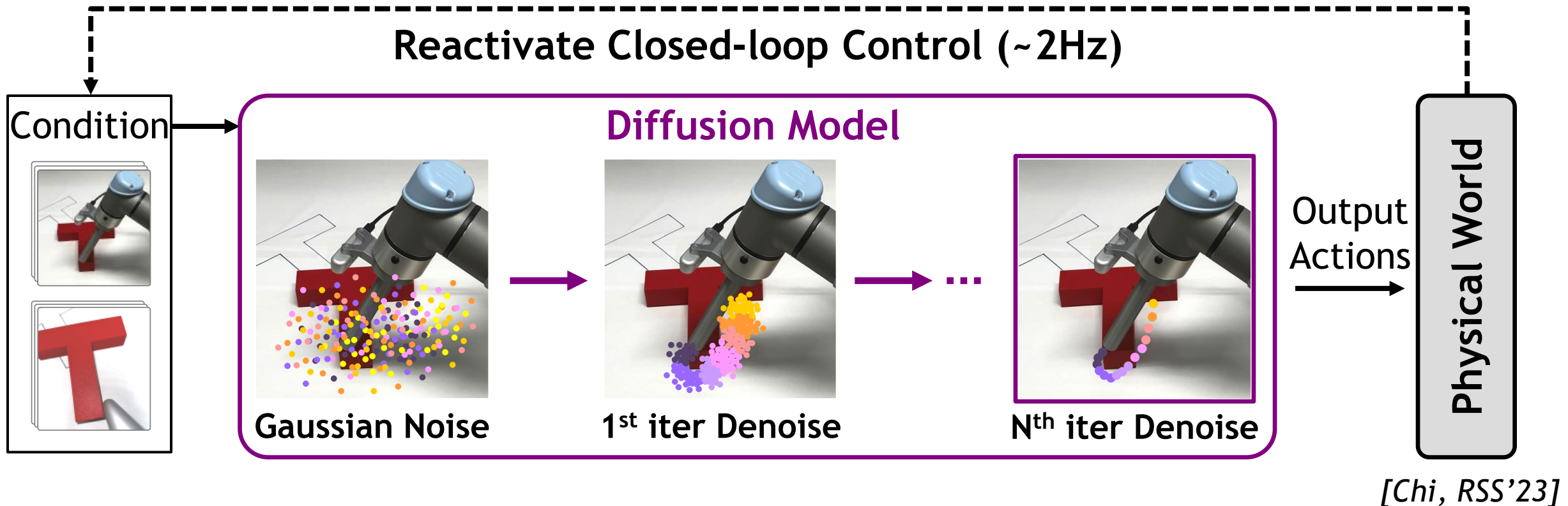
- Emerging methodology for robot manipulation
  - Better temporal action consistency, better training stability...

Inference: **iterative computation** w/ Denoising Diffusion Implicit Model (DDIM)



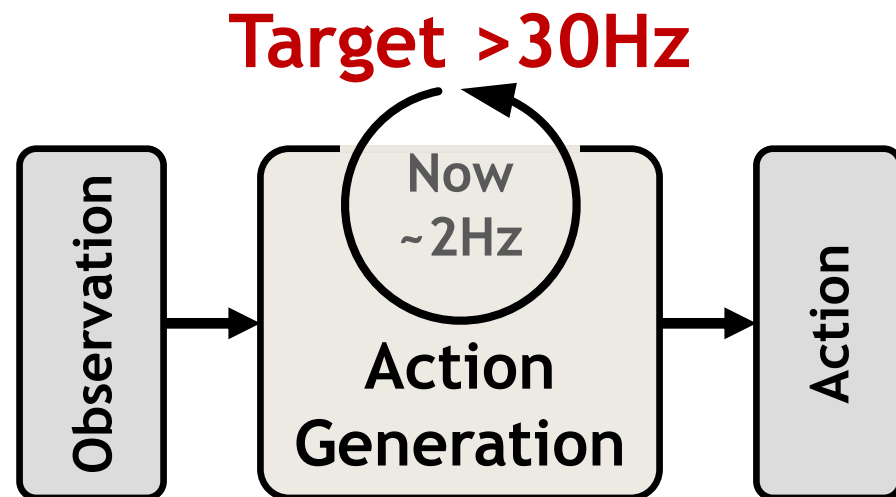
# Diffusion-based Action Generation

- Emerging methodology for robot manipulation
- Deployed with observations or instructions conditioned



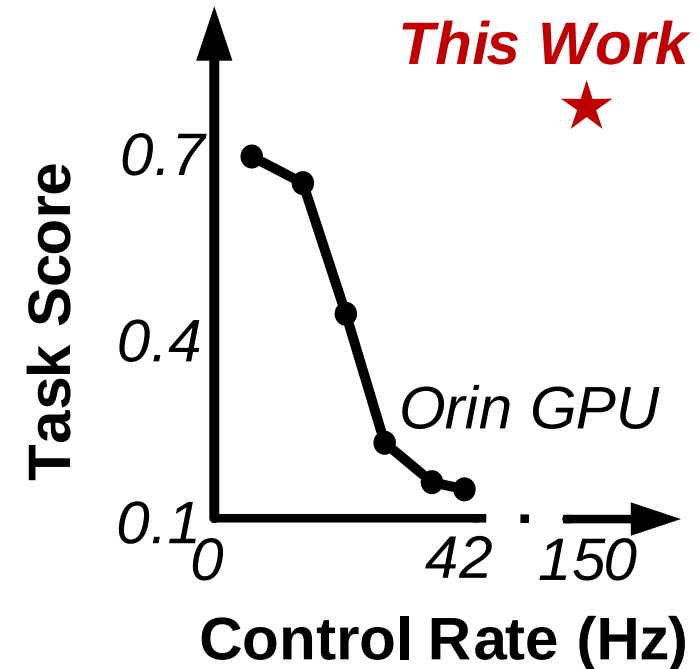
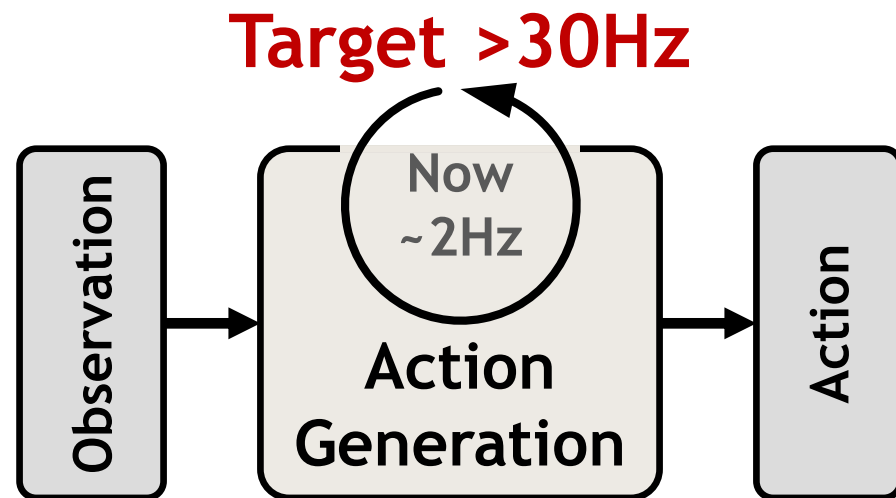
# System Requirements: Low-latency Inference

- Higher control frequency makes more powerful manipulation
  - Faster and smoother actions
  - Rapidly changing scenarios



# System Requirements: Low-latency Inference

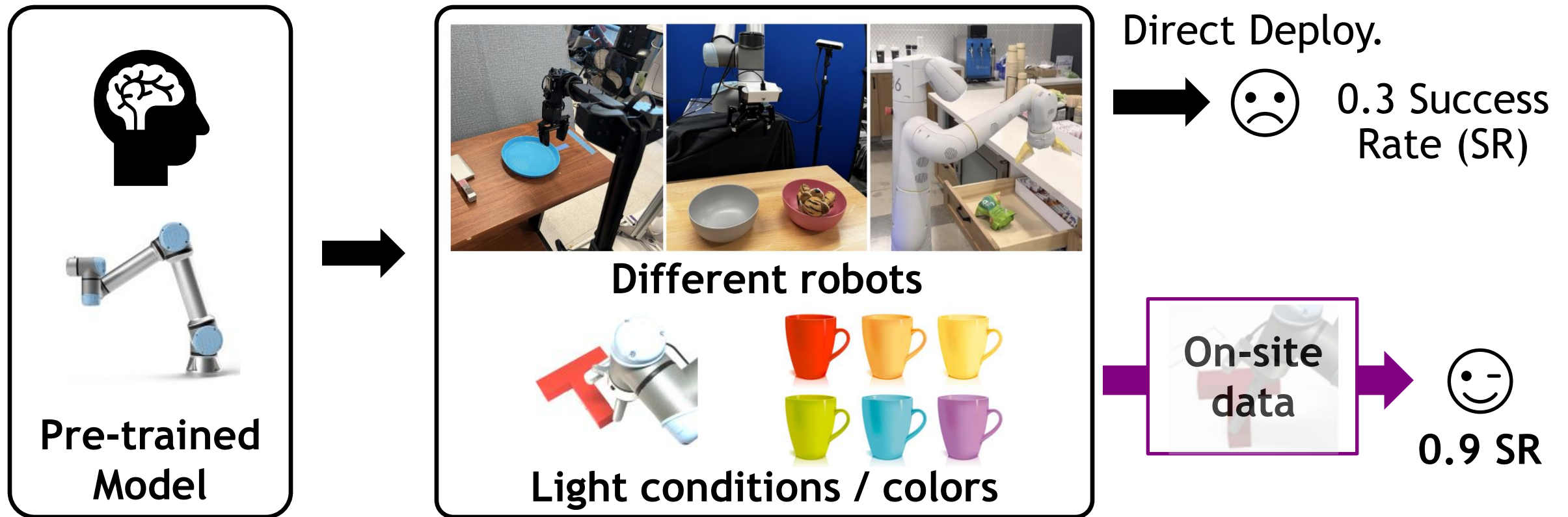
- Higher control frequency makes more powerful manipulation
  - Faster and smoother actions
  - Rapidly changing scenarios
- BUT, diffusion-based AG: too many iterations
  - Reduce denoising steps ?
  - Significant performance loss !





# System Requirements: On-device Fine-tuning

- Adapt to specific environments for better performance
  - Different background, light condition, robot...



# Outline

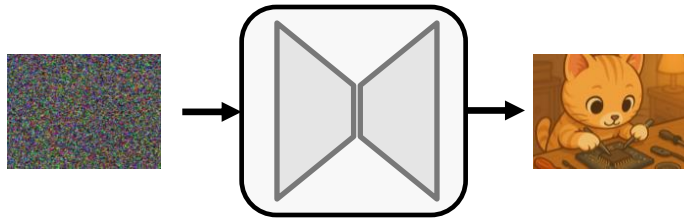
- Introduction
- Challenges of diffusion-transformer-based action generation (DiTAG)
- Edge SoC for DiTAG in robot manipulation
  - Overall architecture
  - Dynamic-time-step speculation-commitment dispatcher (SCD)
  - Differential-input dynamic-bit-width diffusion processor (D3P)
  - On-chip randomness-reduction avoidance
- Measurements and demonstrations
- Conclusion

# Outline

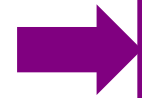
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# Properties of DiT-based Action Gen. (DiTAG)

## Diffusion-based Image Generation

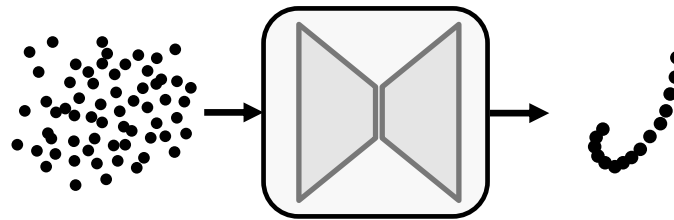


Sequence length:  $\sim 10^2$

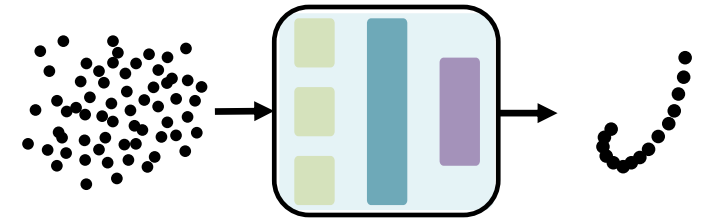


## Diffusion-based Action Generation

### UNet-based



### DiTAG



Sequence length: 8~16, minimal data reuse

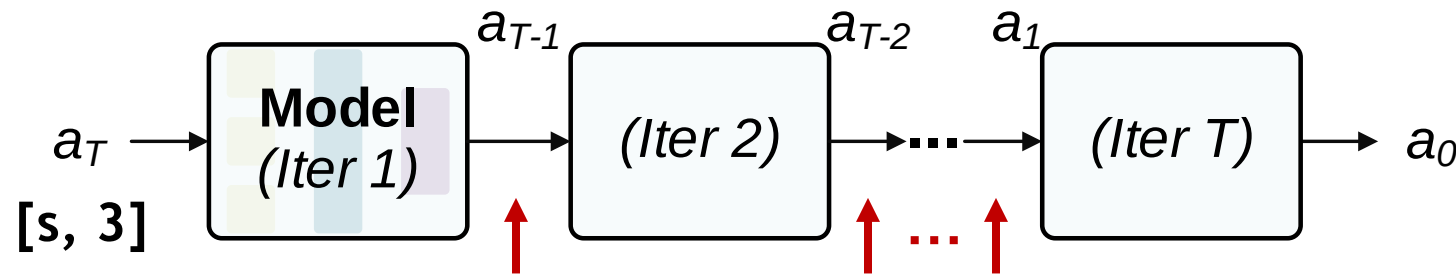
1D Convolution layers

Linear / Attention layers

10x less data reuse w/ better manipulation

# Inference Challenge: Strong Data Dependency

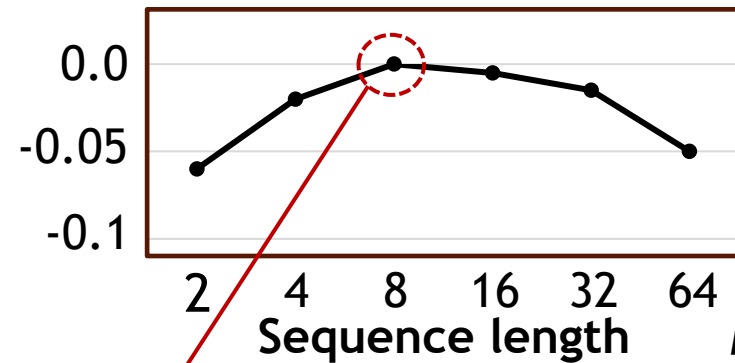
- DiTAG has a deep sequential process w/ small input
  - Single batch, limited sequence length



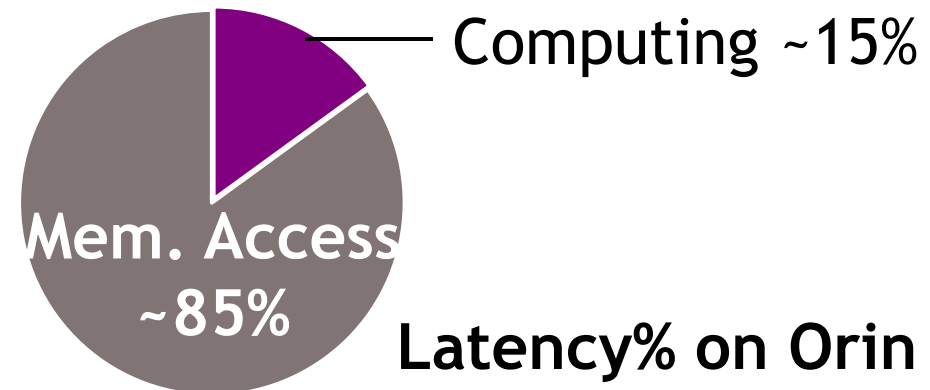
**Weights Re-loading**

**Results in a severe  
memory-access bottleneck**

Relative Performance loss



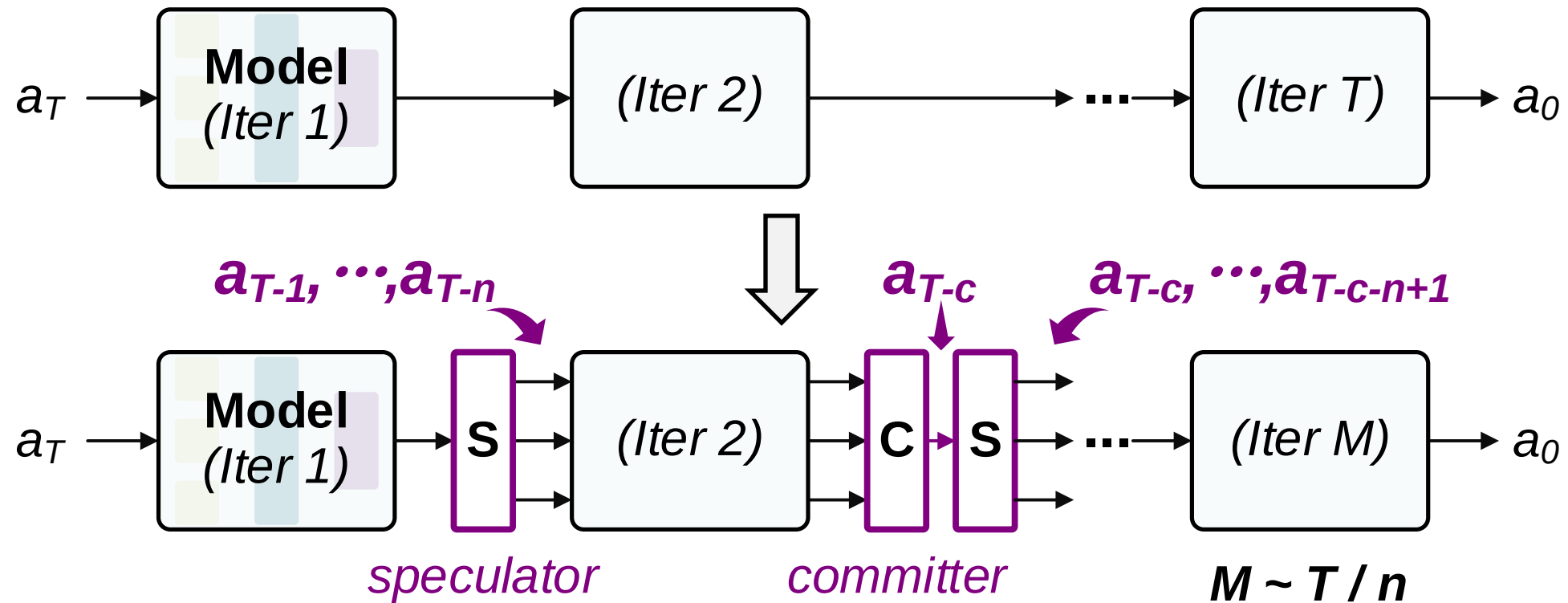
[Chi, RSS'23]



Latency% on Orin

# Solution: Speculative Denoising

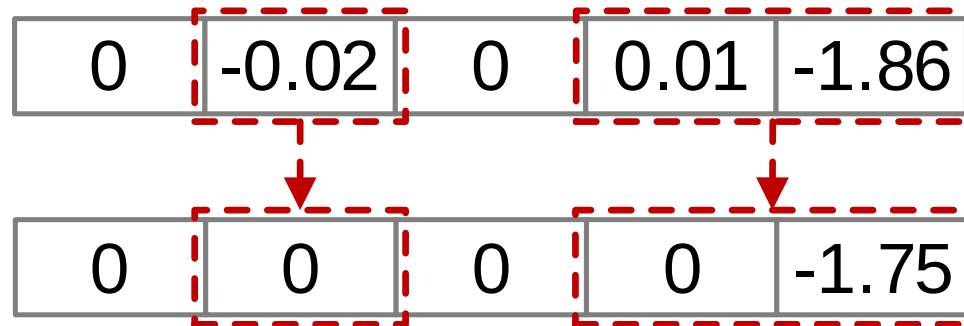
- Time-step-varying speculation: sequential  $\rightarrow$  parallel
  - Increased batch size, less iterations  $\rightarrow$  improved data reuse and utilization



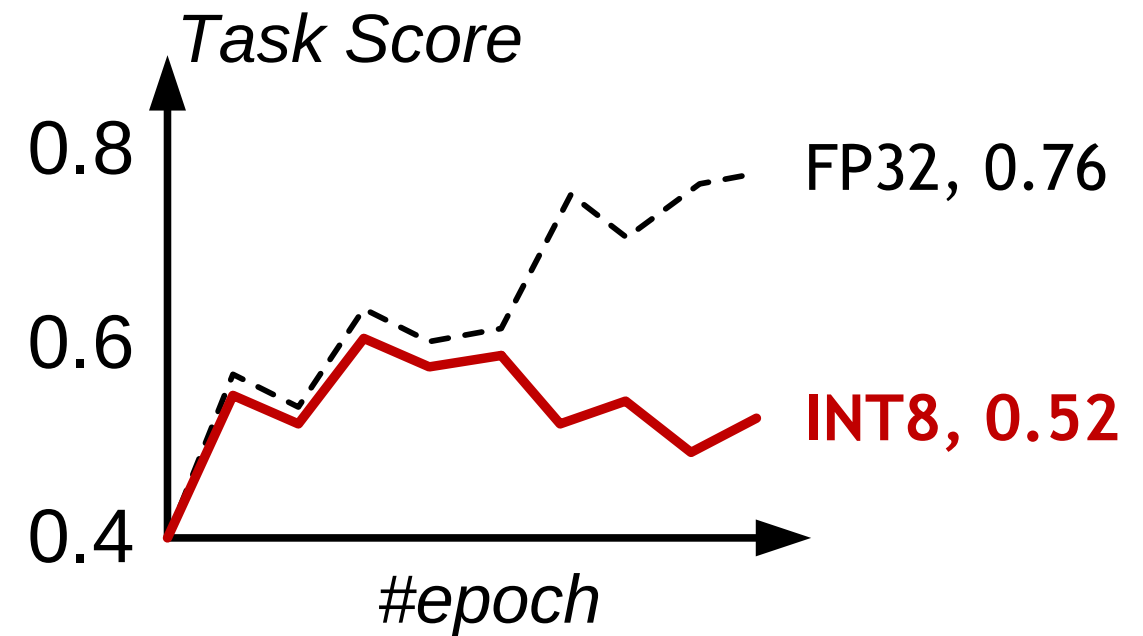
# Fine-tune Challenge: Randomness Reduction

- **Low-bit fine-tune on edge: lower cost, worse performance**
  - Quantization induces randomness reduction → important to diffusion training

Quantized Gradients

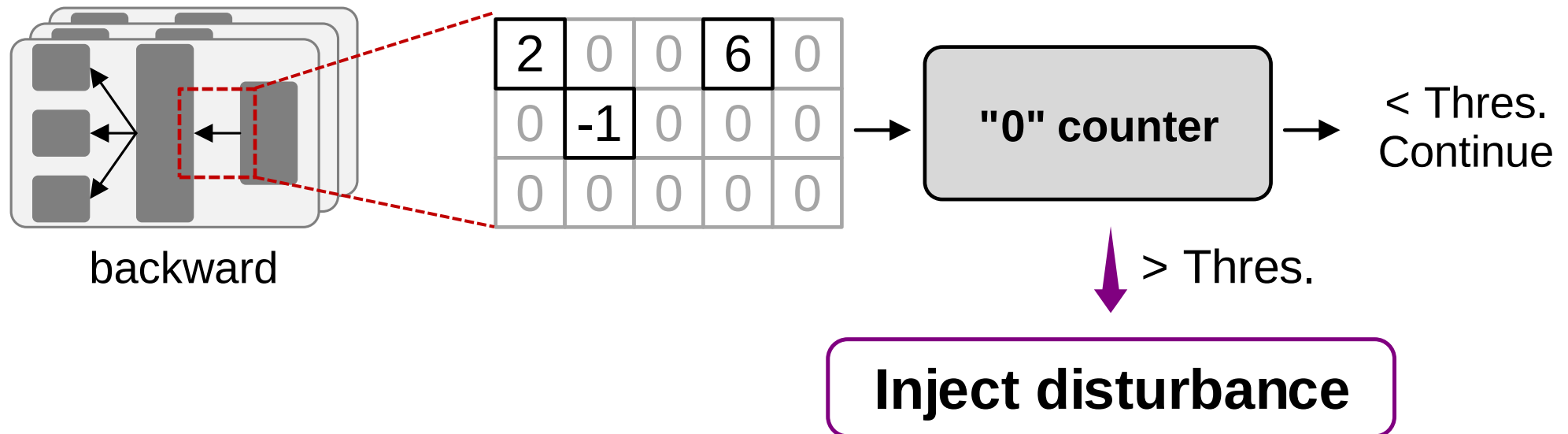


Results in uneven denoising performance across timesteps



# Solution: On-chip Disturbance Enhancing

- On-chip randomness reduction detection
  - “Timestep overfitting” → Gradient vanishing, especially in low-bit fine-tune
- Increase the random timestep interval → Enhance disturbance

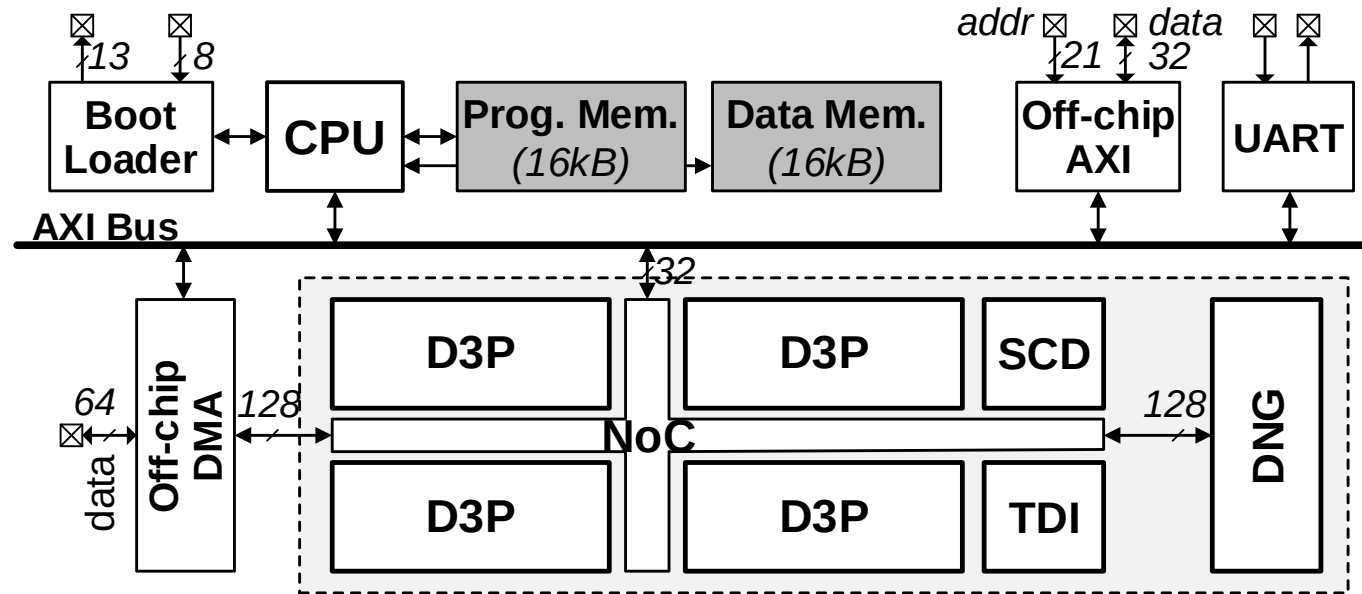




# Outline

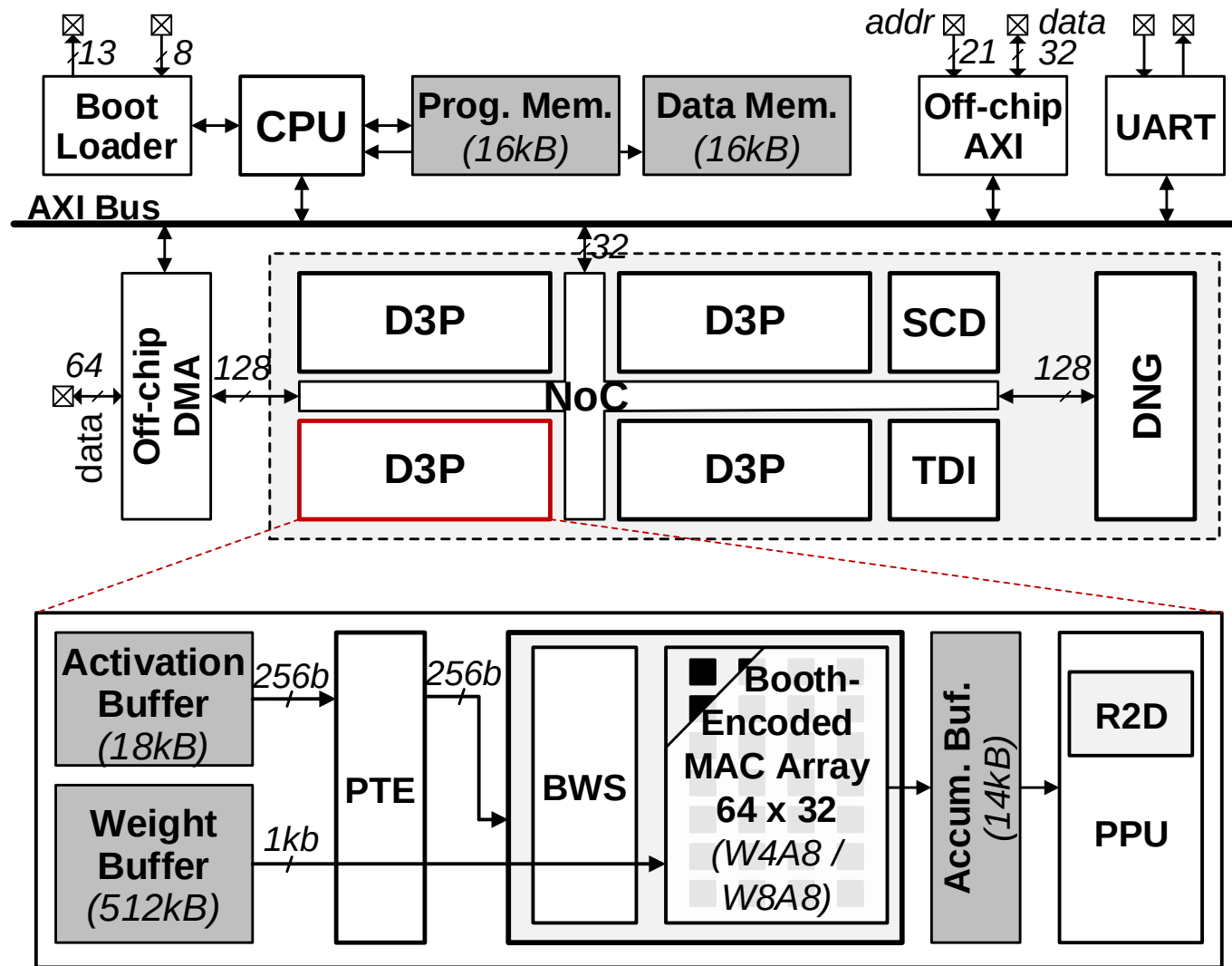
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- Conclusion

# Architecture Overview



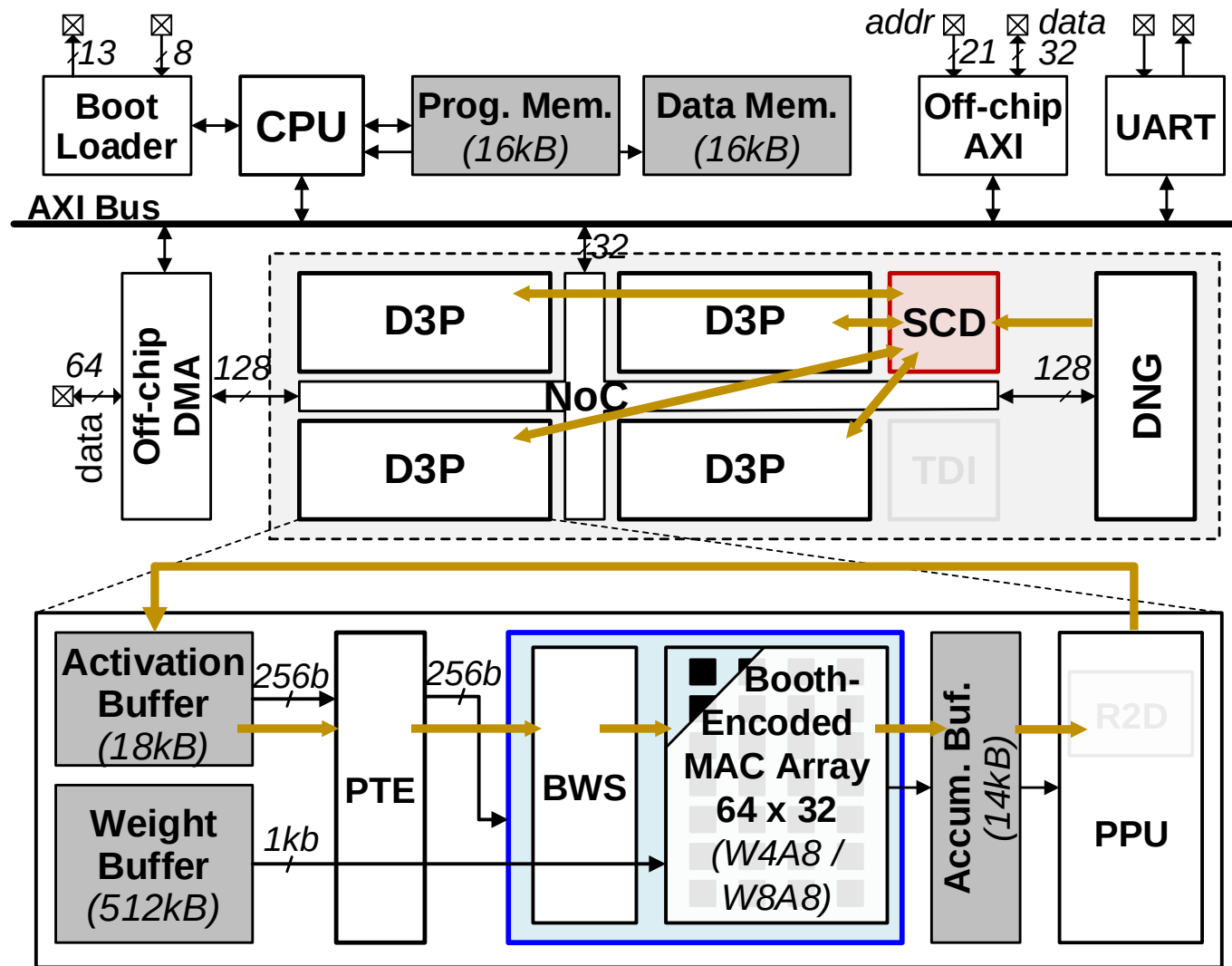
- Demonstrate 4-core DiTAG accelerator
- Dual-mode noise generator (DNG)
- RISC-V CPU for program
- Off-chip DMA for weights loading

# Architecture Overview



- Demonstrate 4-core DiTAG accelerator
- Dual-mode noise generator (DNG)
- RISC-V CPU for program
- Off-chip DMA for weights loading
- Local buffering of act. and weights
- preprocess-and-transpose engine (PTE) for on-the-fly input processing
- $64 \times 32$  booth-encoded MAC array, supporting weights in INT8/INT4/FP4 with bit-width selection (BWS)

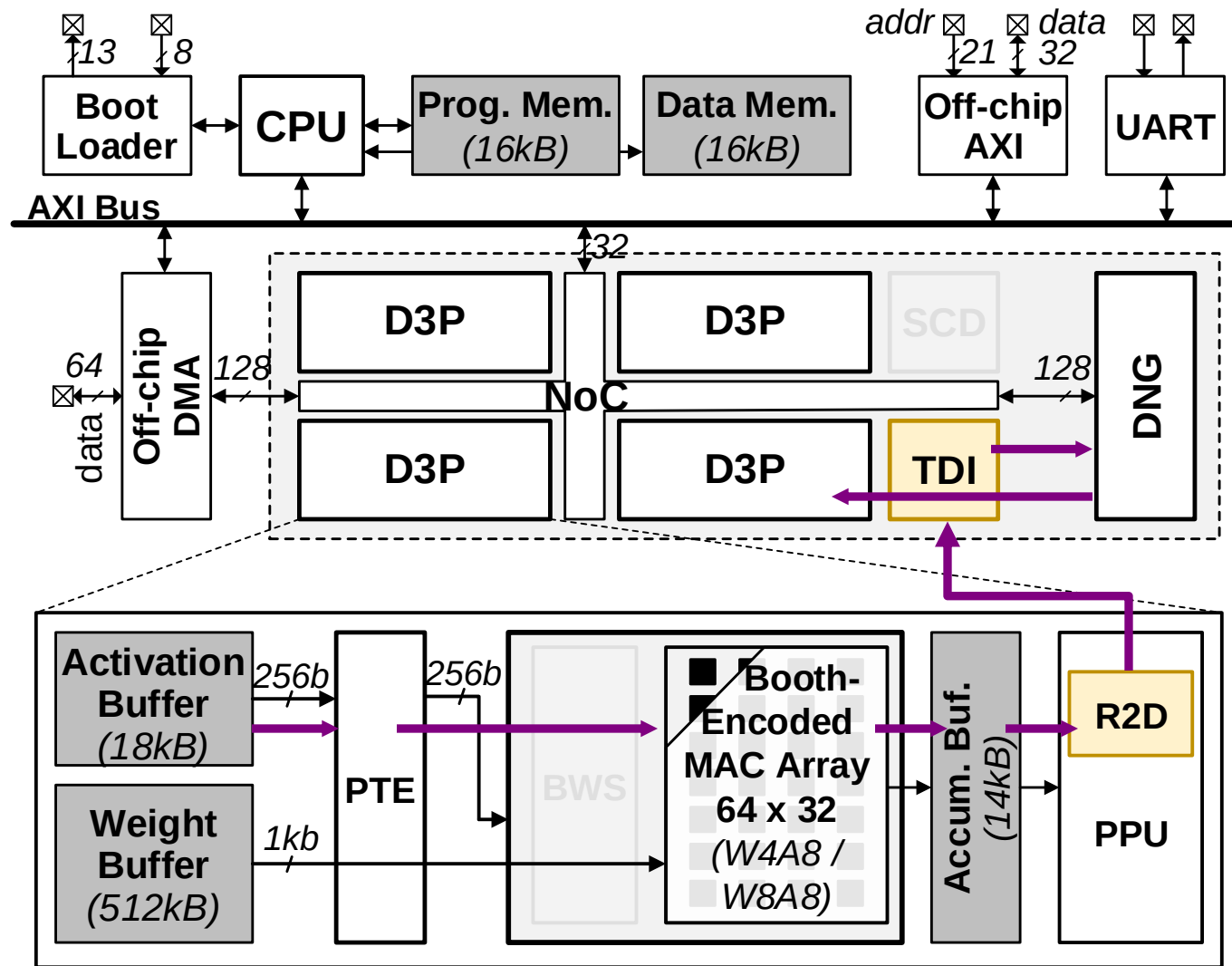
# Architecture Overview: Inference Flow



**Feature 1: SCD for dynamic speculative parallel/skip HW dispatching**

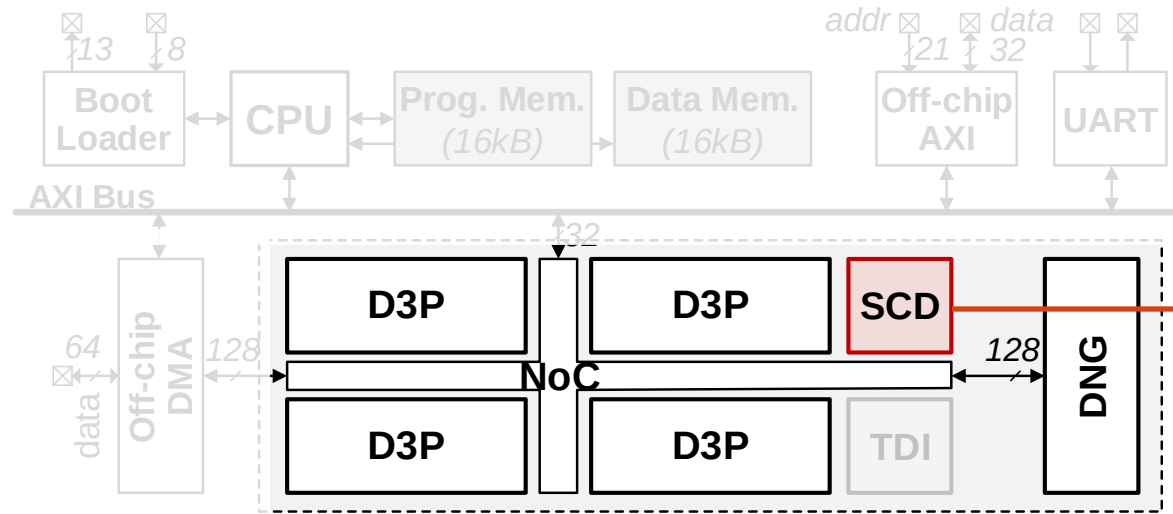
**Feature 2: BWS and MAC array for efficient differential-input denoising with dynamic-bit-width**

# Architecture Overview: Fine-tune Flow



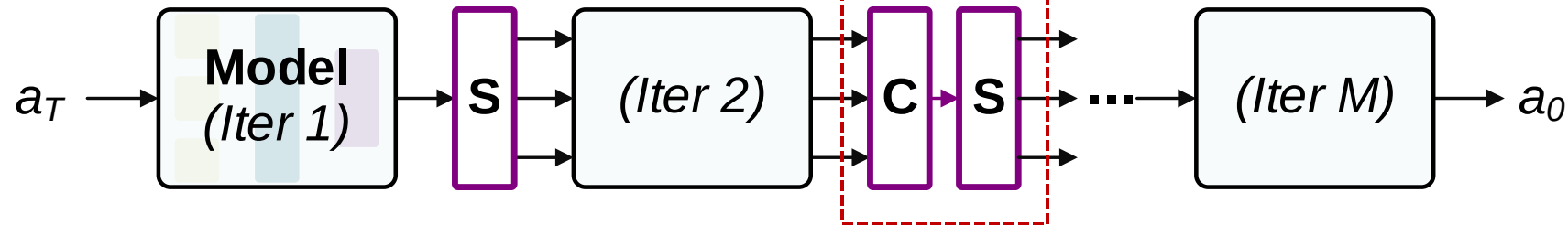
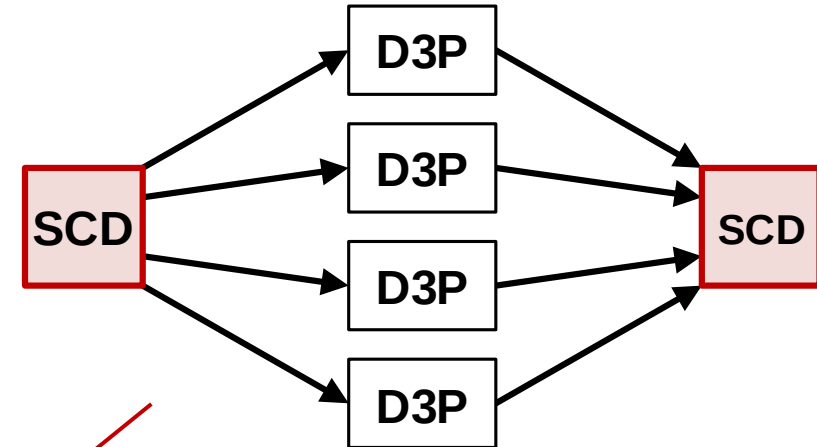
**Feature 3: Timestep-disturbance injector (TDI) and randomness-reduction detector (R2D) stabilizing fine-tune accuracy**

# Trade Comput. for Mem. Access: Speculation



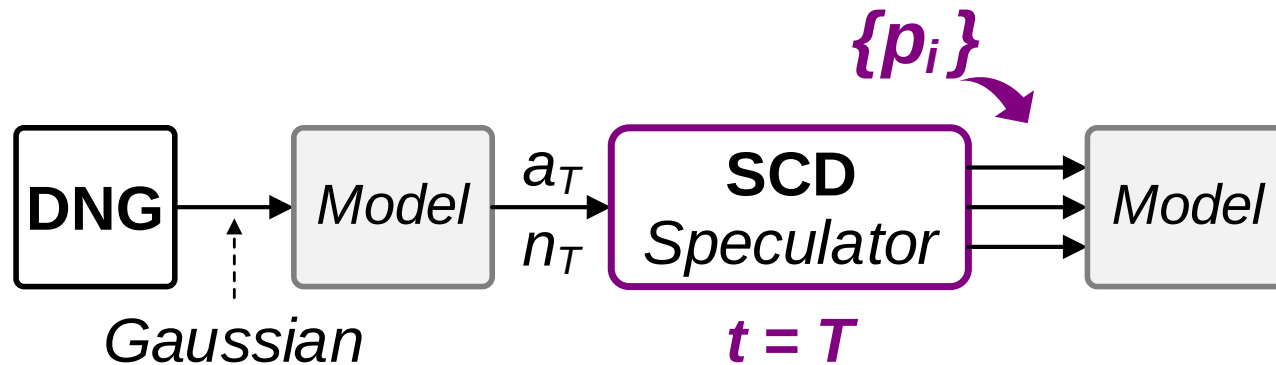
Speculation

Commit

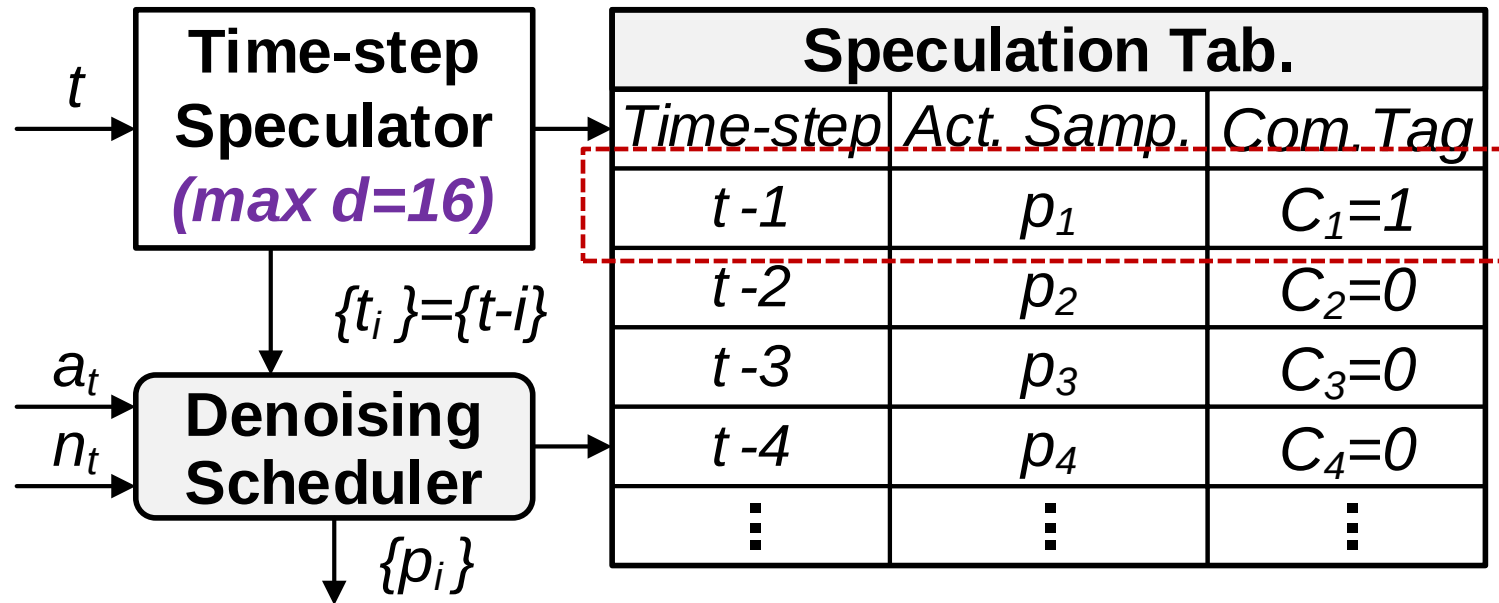


Sequential process w/ strong dependency  $\rightarrow$  partially parallel process

# Dynamic Dispatcher: Speculation

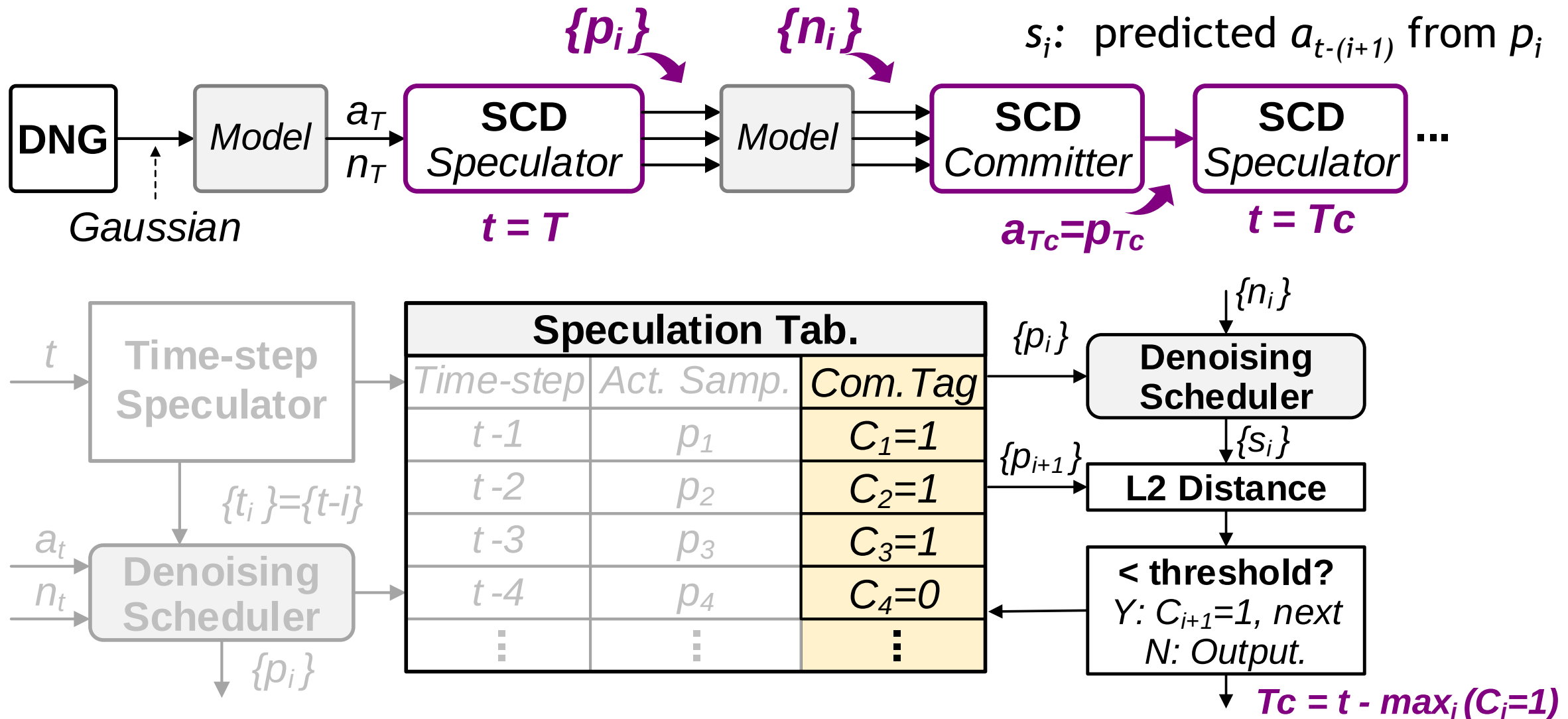


$p_i$ : predicted  $a_{t-i}$   
 $n_i$ : predicted noise of  $t_i$   
 $d$ : speculation depth



Worst case  
 No redundant iterations

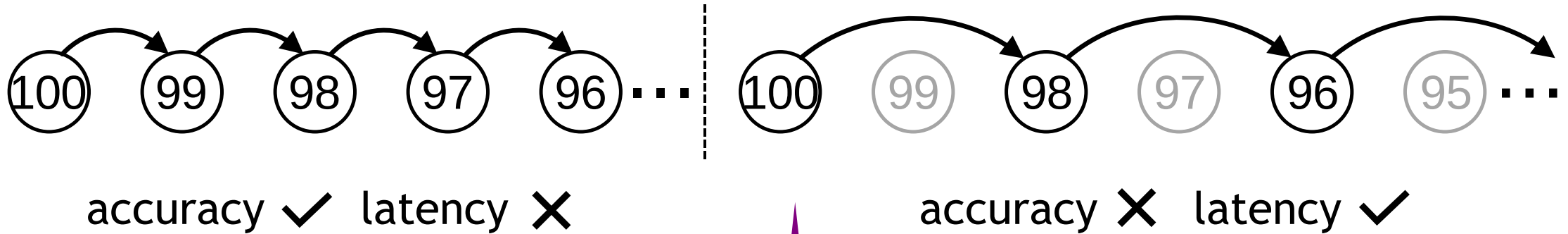
# Dynamic Dispatcher: Commit





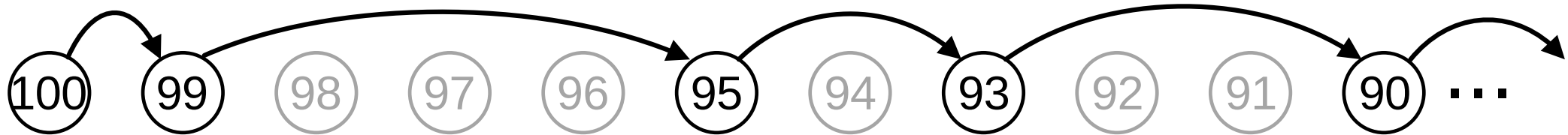
# Results: Optimized Denoise Iterations

Conventional diffusion process w/ uniform timestep interval



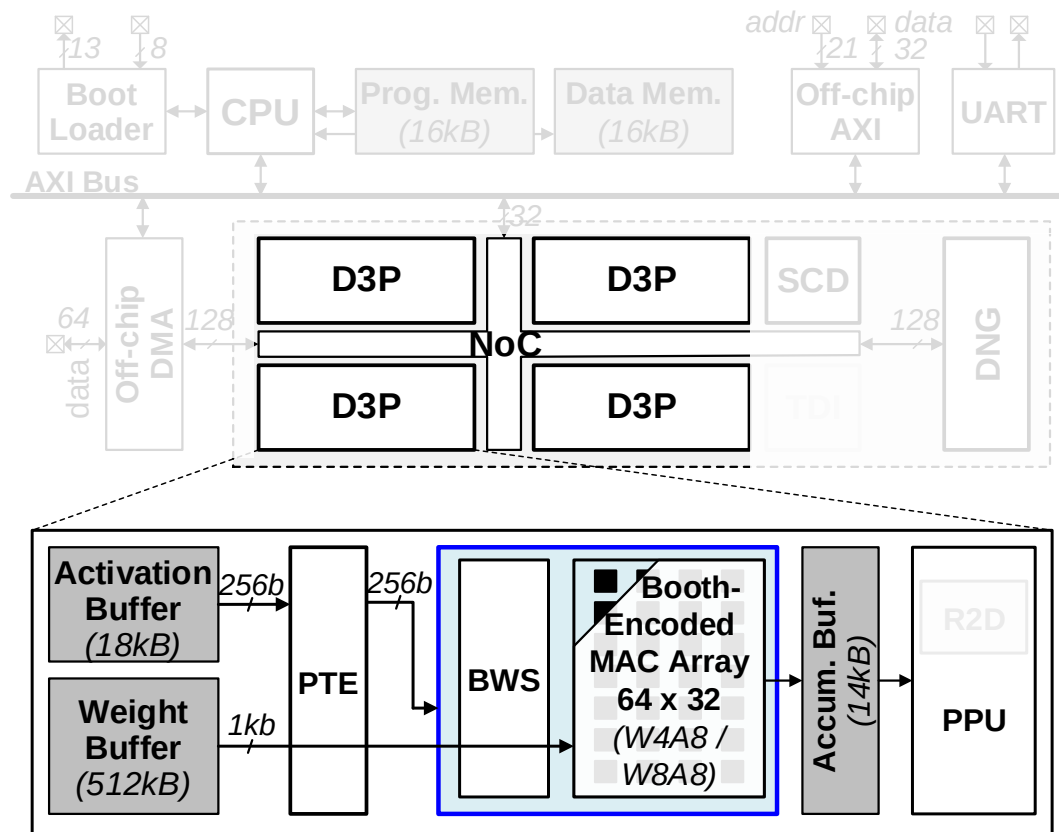
**Speculative-Commit Dispatcher**

This work



**Skip redundant iterations, accuracy ✓ latency ✓**

# Further Acceleration: D3P



## Normalized Latency



↓ *SCD: d=8, step=16*



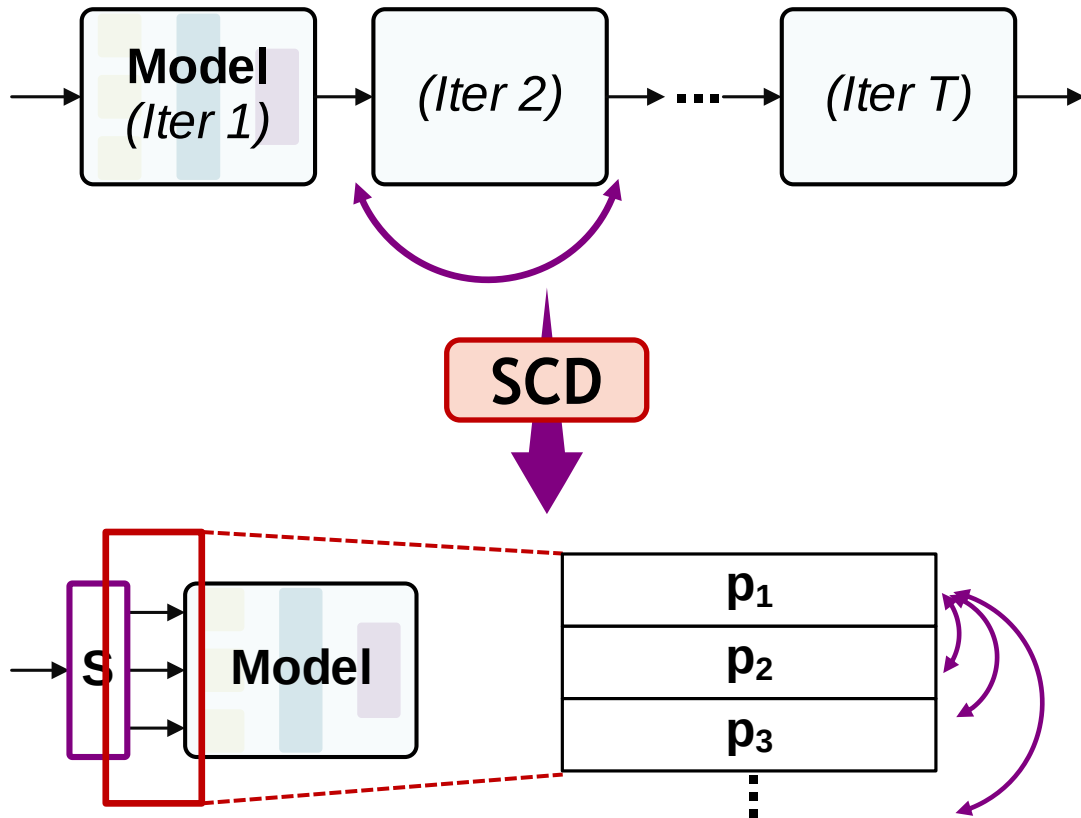
↓ *D3P*



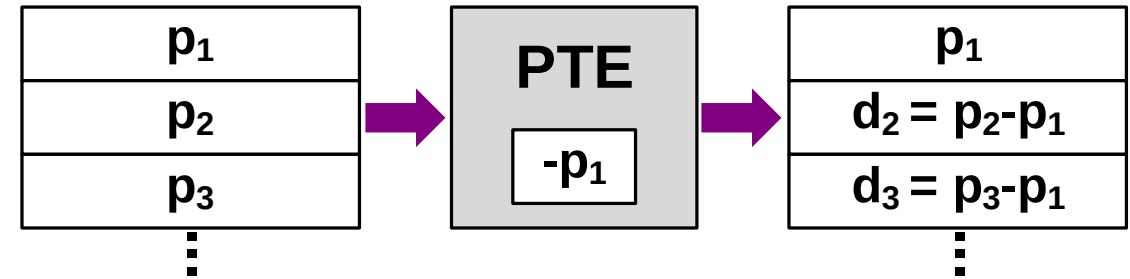
Computation increased by speculation → Reduce computing latency

# Intra-Input Similarity Introduced by SCD

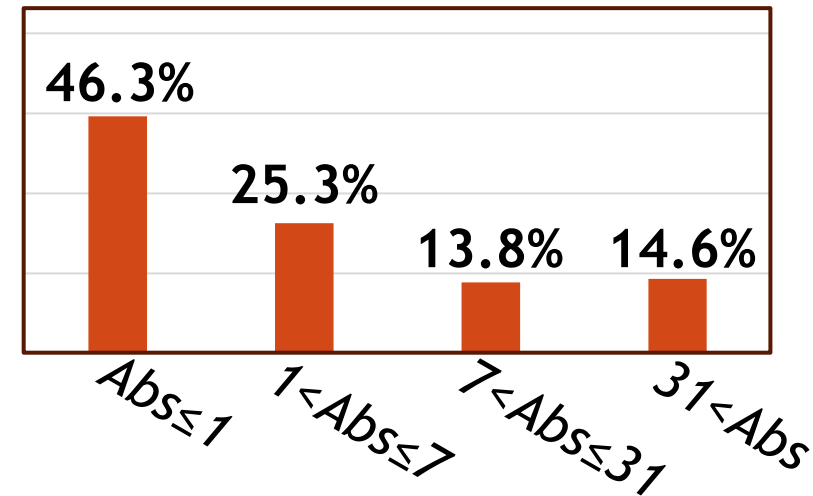
- Inter-time similarity  $\rightarrow$  **intra-input similarity**



On-the-fly Differential Process



Distribution of input data (quantized)

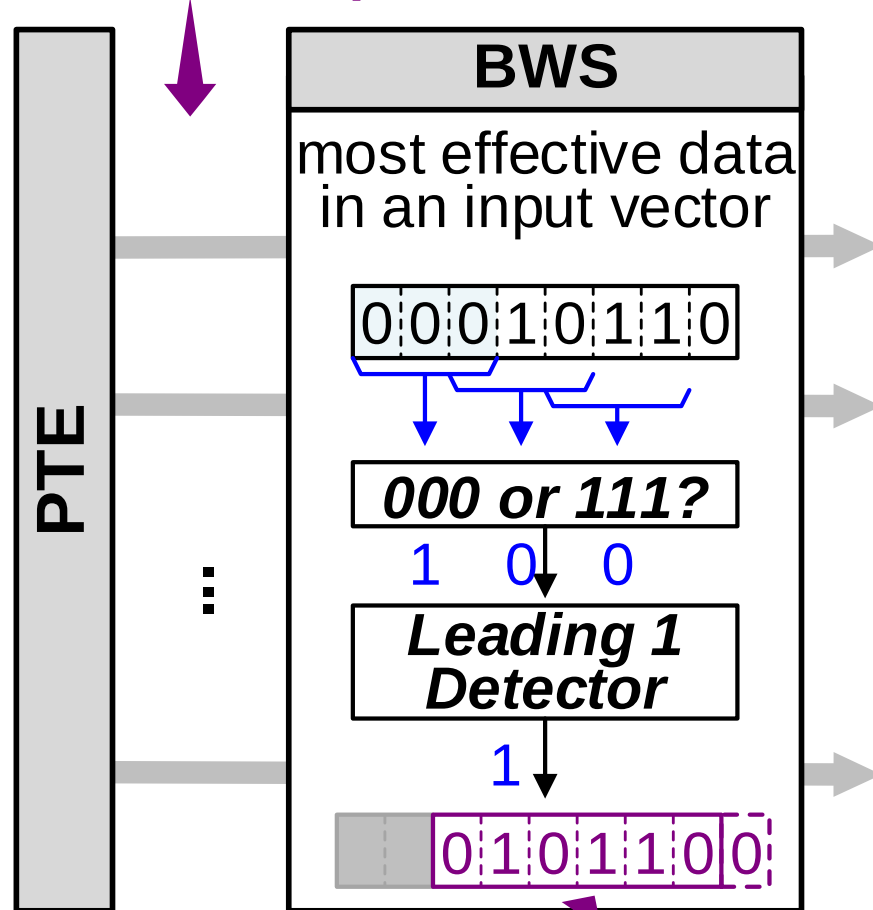


**Bit-level Redundancy**

*Measured on Push-T, at feedforward layer in the 1<sup>st</sup> transformer block*

# Dynamic-bit-width Differential-input MAC

Differential input

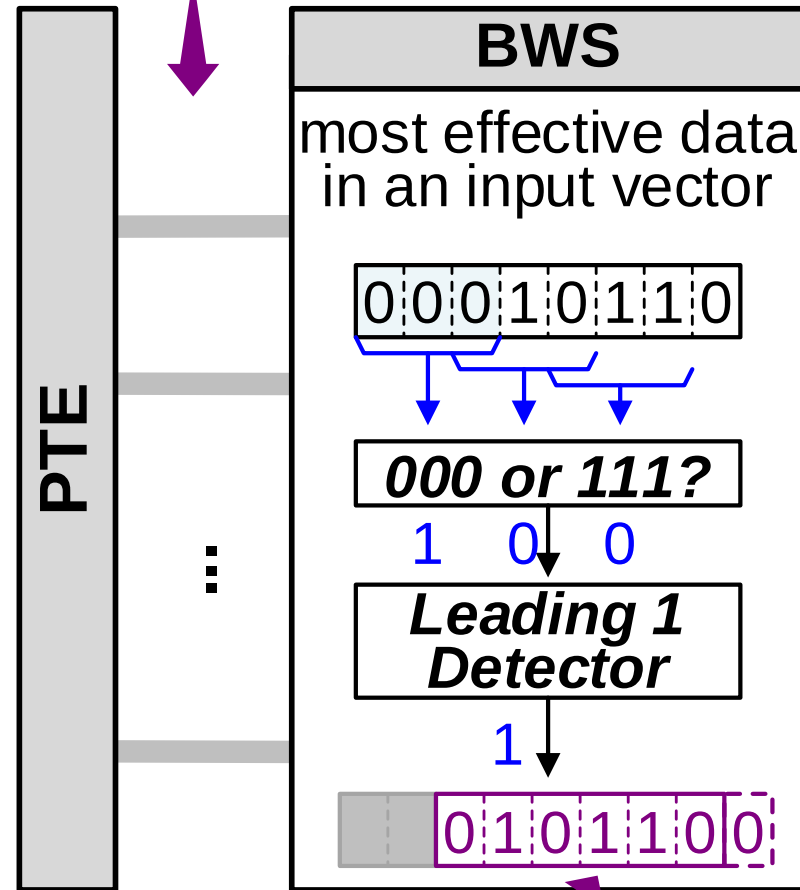


To be computed

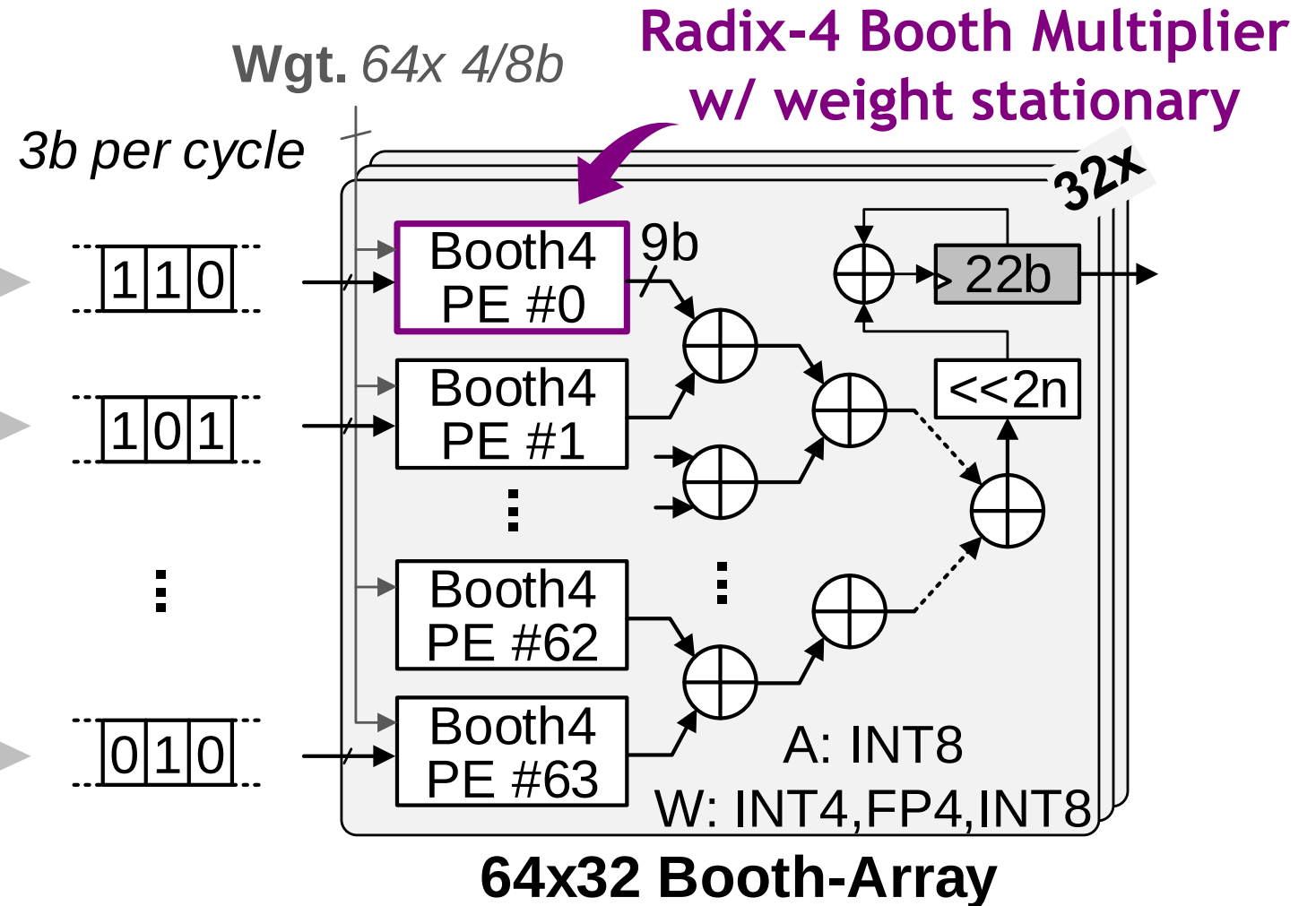
BWS predict effective bit-width of input for following MAC array

# Dynamic-bit-width Differential-input MAC

Differential input

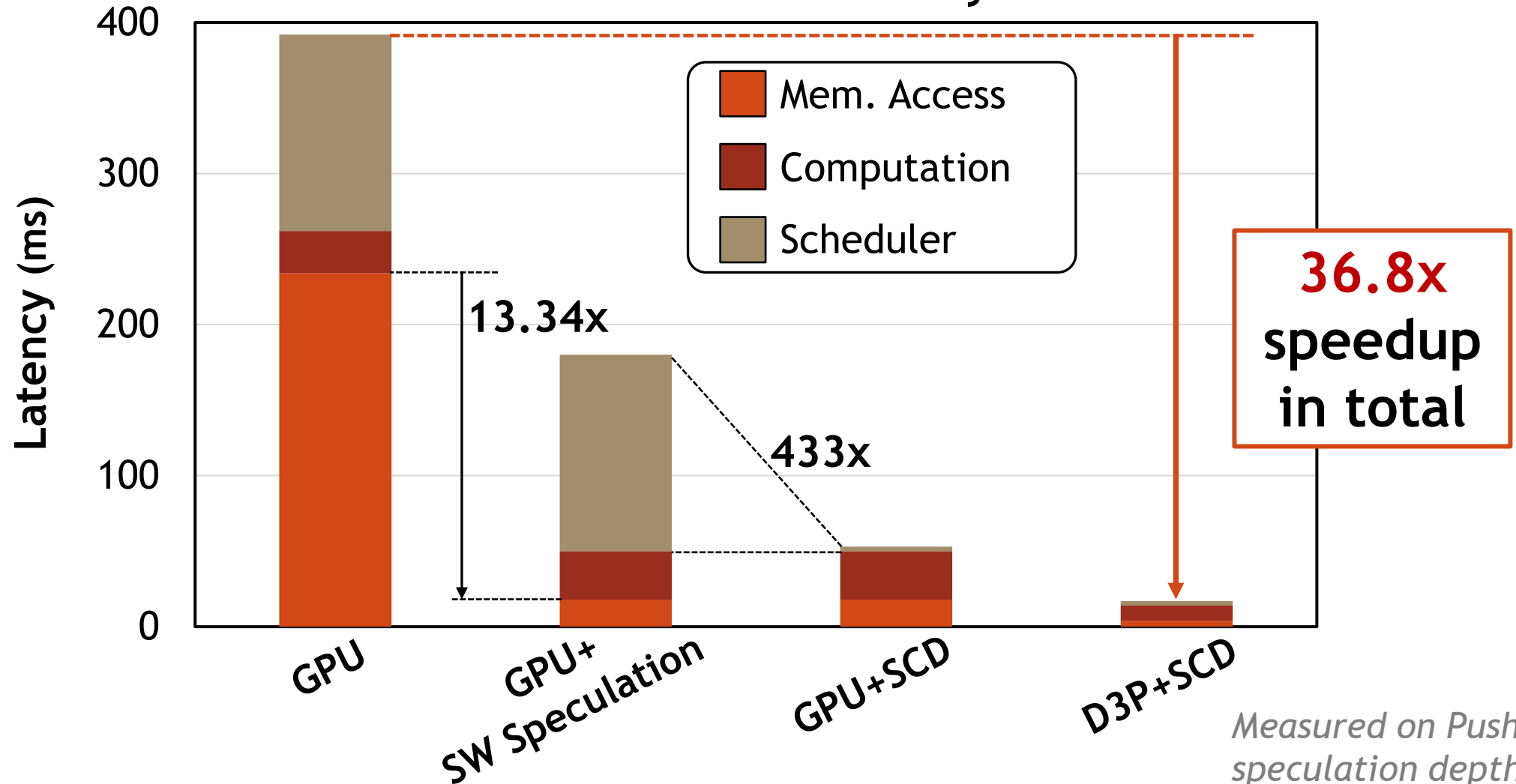


To be computed

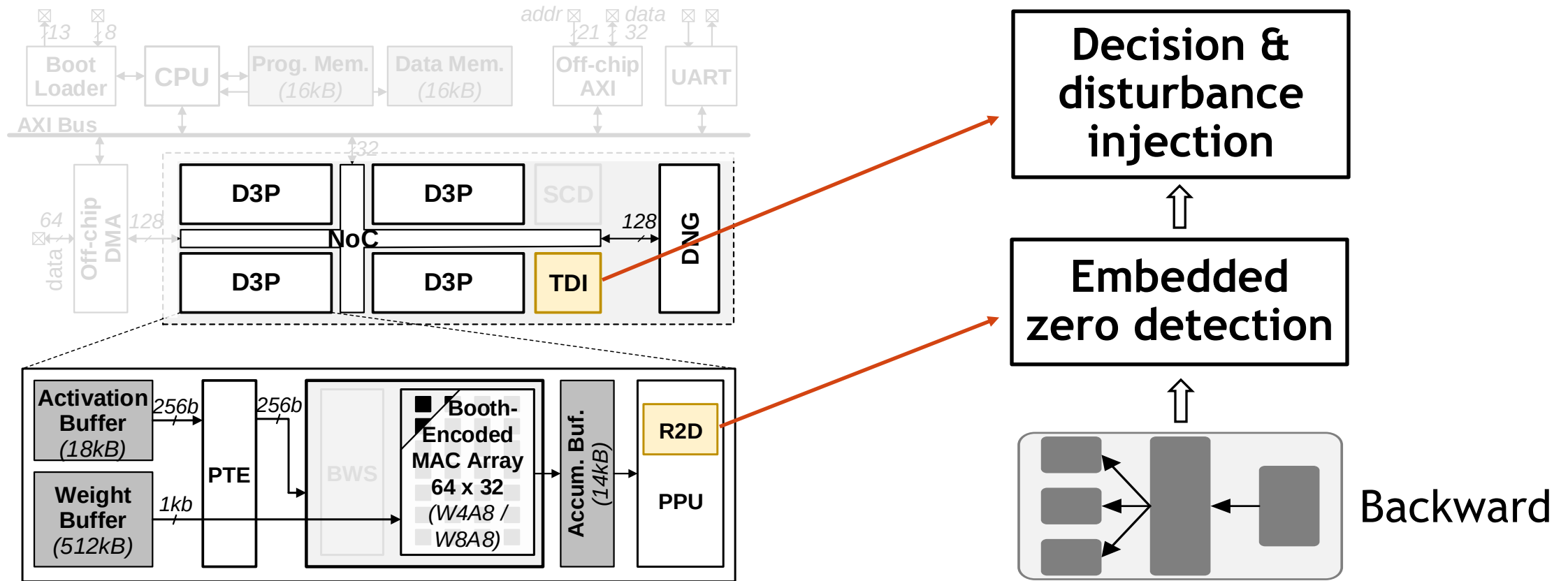


# Improved Inference Latency Breakdown

## INT8 Inference Latency



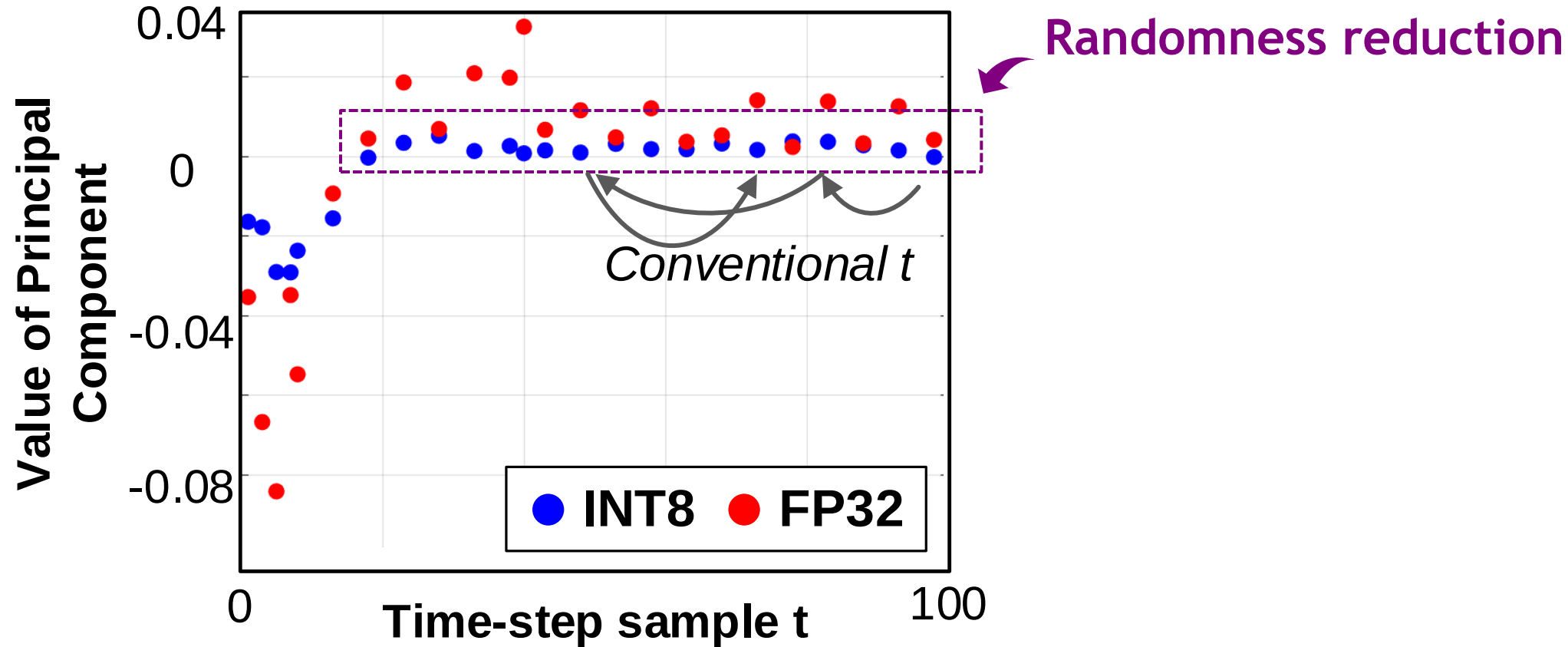
# Support Low-bit Fine-tune: TDI & R2D



Randomness reduction → Low-cost disturbance enhancement

# Quantization-induced Randomness Reduction

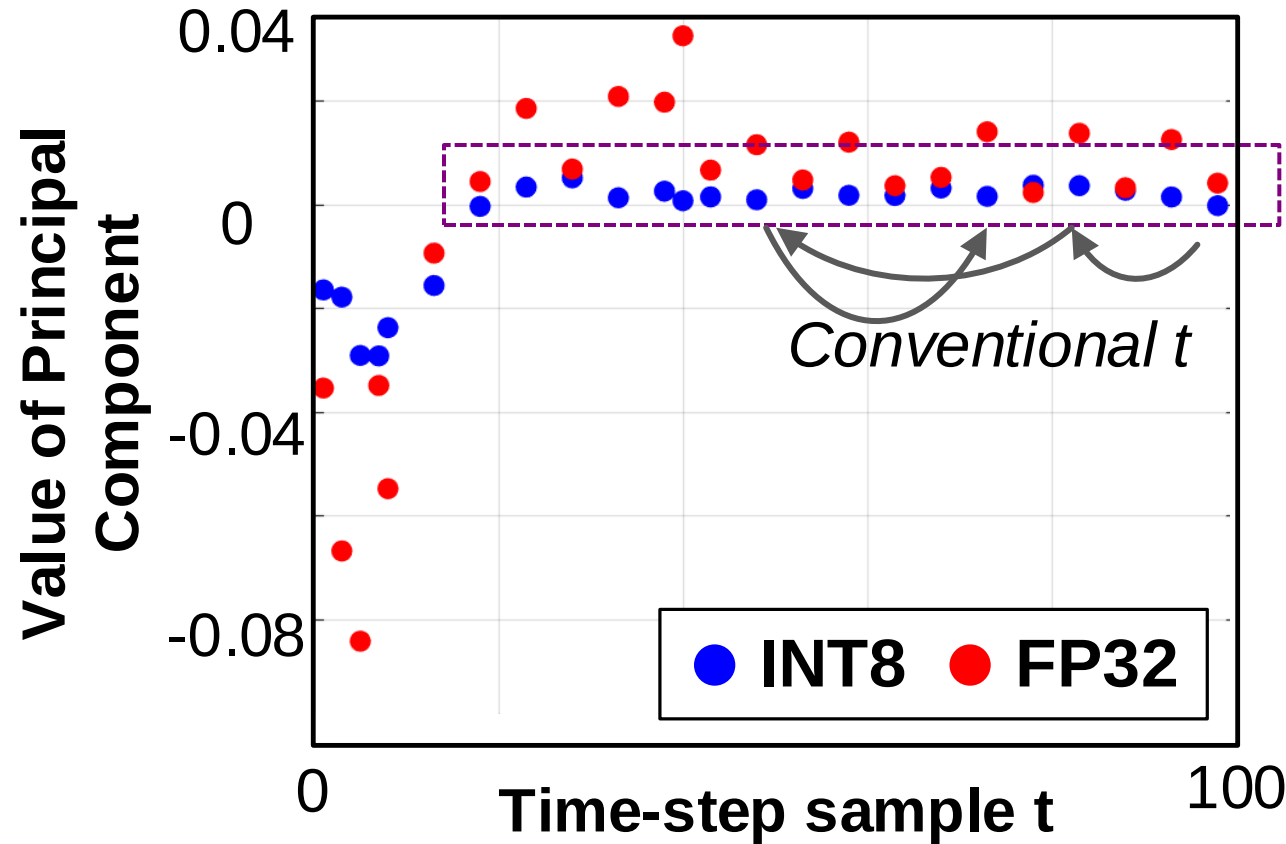
Gradient principal components over timesteps



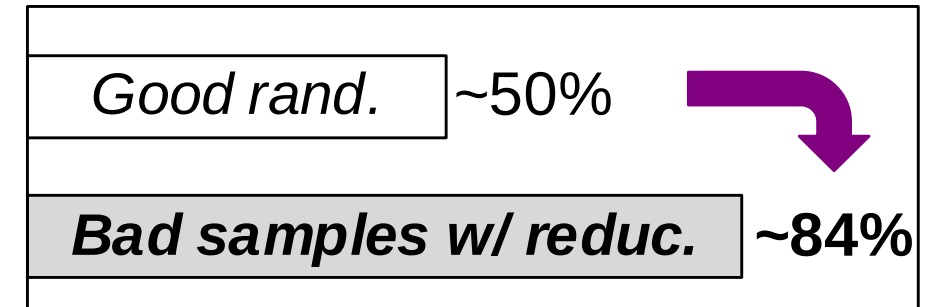


# Quantization-induced Randomness Reduction

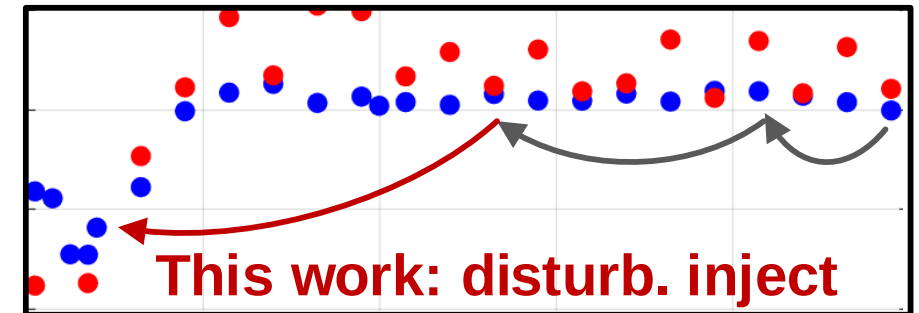
Gradient principal components over timesteps



Zero-gradient ratio



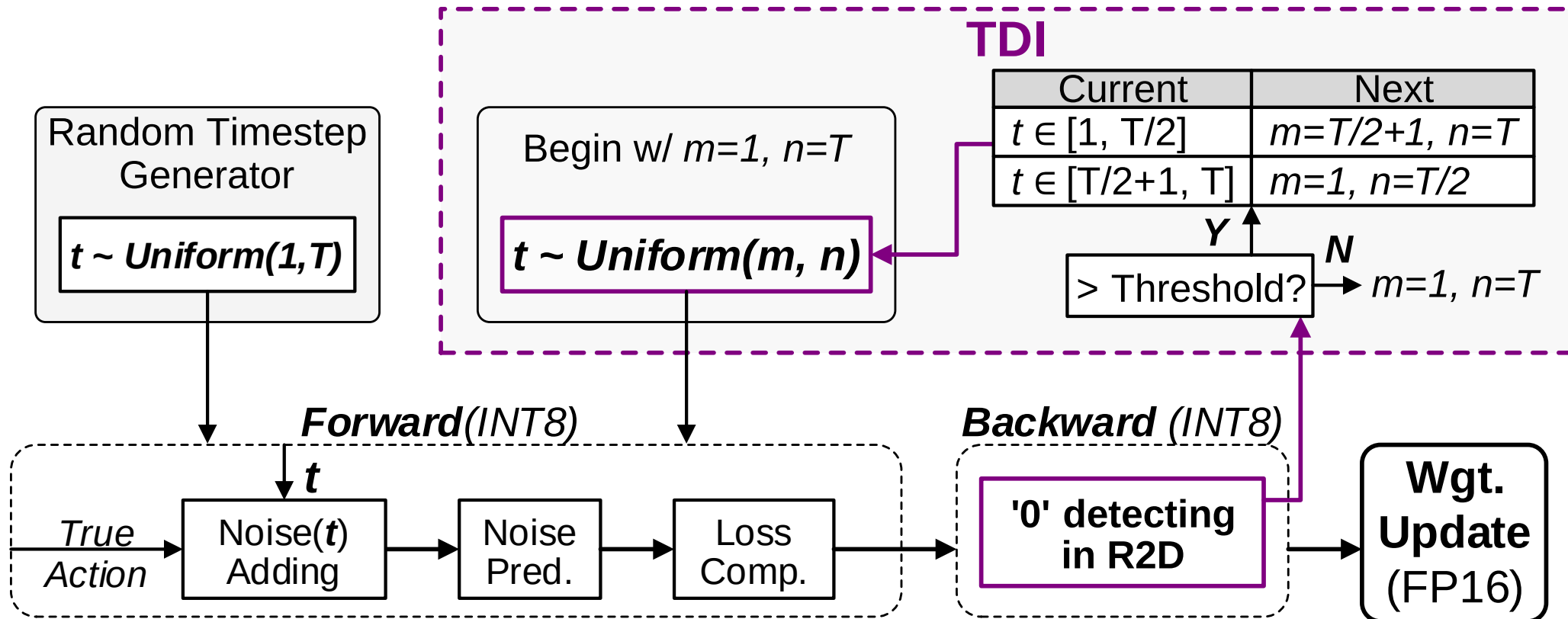
TDI + R2D



Achieve optimized timestep

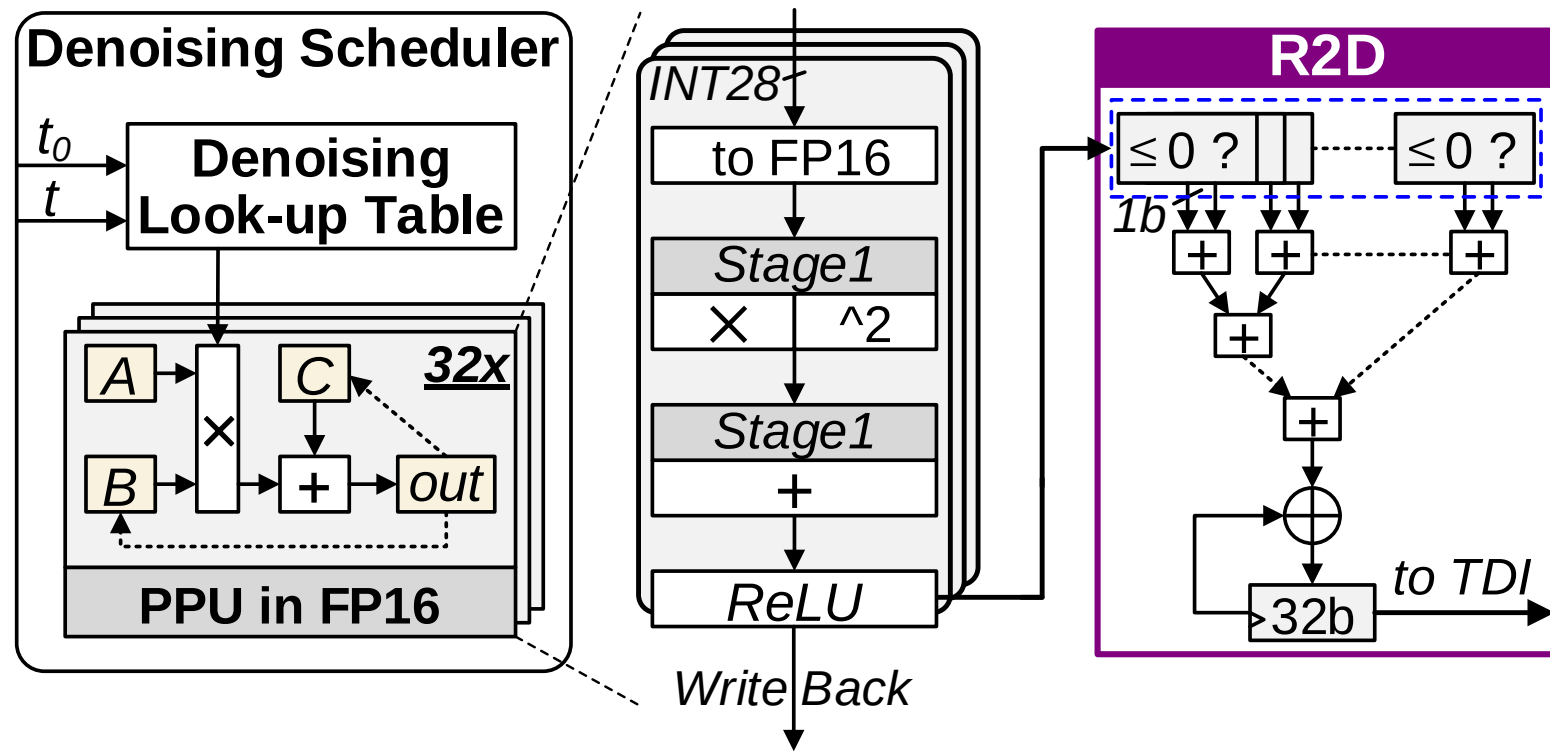
# Disturbance-Injected Fine-tune Flow

- Conventional random timestep generation: independent
- Disturbance-injected flow:  $t$  generation controlled by gradients

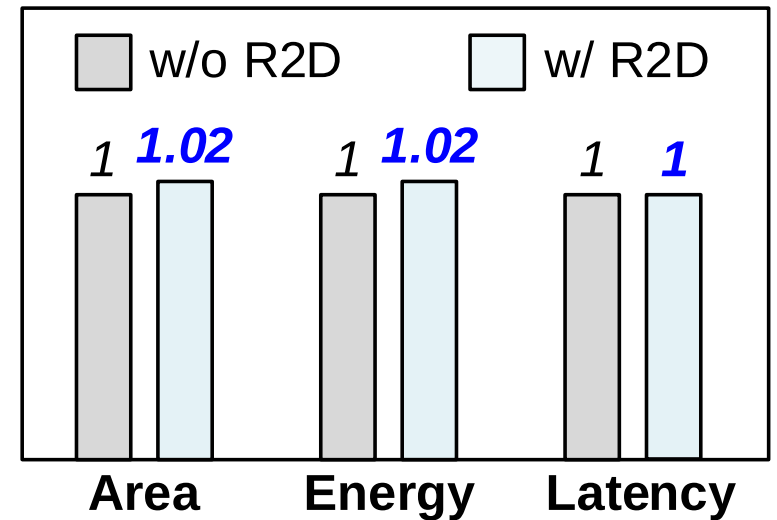


# Programmable PPU & Embedded R2D

- Support linear/non-linear operators: denoising scheduler, layernorm...
- R2D reuses ReLU units w/ 1-bit-input adder tree



Normalized Overhead of PPU

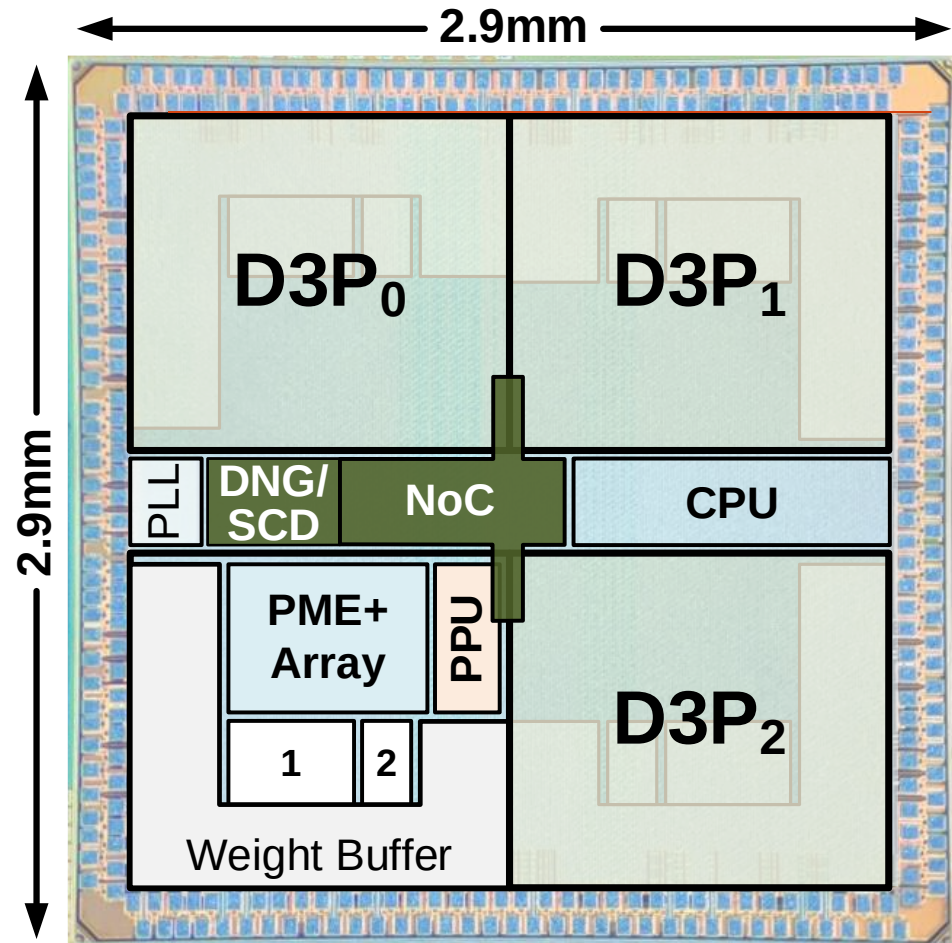


**Minimal overhead**

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  - Differential-input dynamic-bit-width diffusion processor (D3P)
  - On-chip randomness-reduction avoidance
- **Measurements and demonstrations**
- Conclusion

# Prototype Measurements

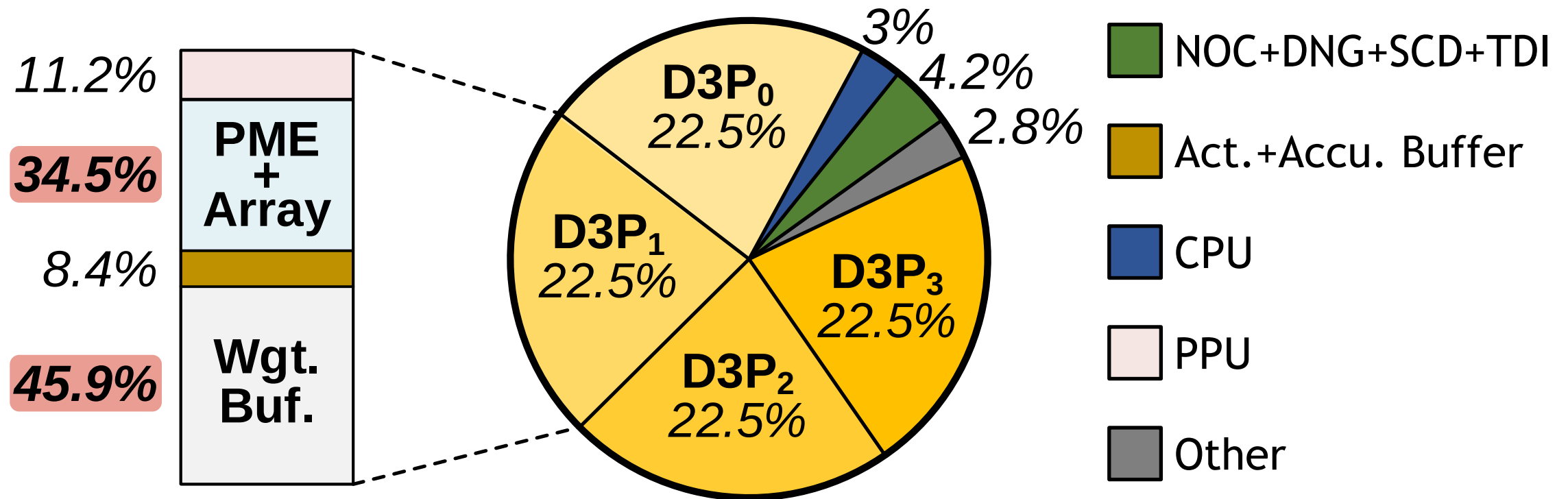


**1** Activation Buffer   **2** Accumulation Buffer

| Chip Summary                         |                                      |
|--------------------------------------|--------------------------------------|
| Technology                           | 28nm CMOS                            |
| Chip/Core Area                       | 8.41/6.25 mm <sup>2</sup>            |
| Application                          | Robot Manipulation Action Generation |
| On-chip SRAM                         | 2208 kB                              |
| Supply Voltage                       | 0.6-1.2 V                            |
| Clock Frequency                      | 250-1250 MHz                         |
| Data Precision                       | W: INT4/FP4/INT8, A: INT8            |
| 8b Throughput <sup>1</sup>           | 6.4 TOPS                             |
| Inference Latency <sup>1,2</sup>     | 10.6 ms                              |
| Inference Energy Eff. <sup>1,2</sup> | 8.6 mJ/infer.                        |
| Fine-tune Energy Eff. <sup>1,3</sup> | 7.4 mJ/epoch                         |

1. Measured at 0.9V, 800MHz, Push-T task. 2. Fully on chip.  
3. 1 epoch = 8 steps w/ a batch size of 16

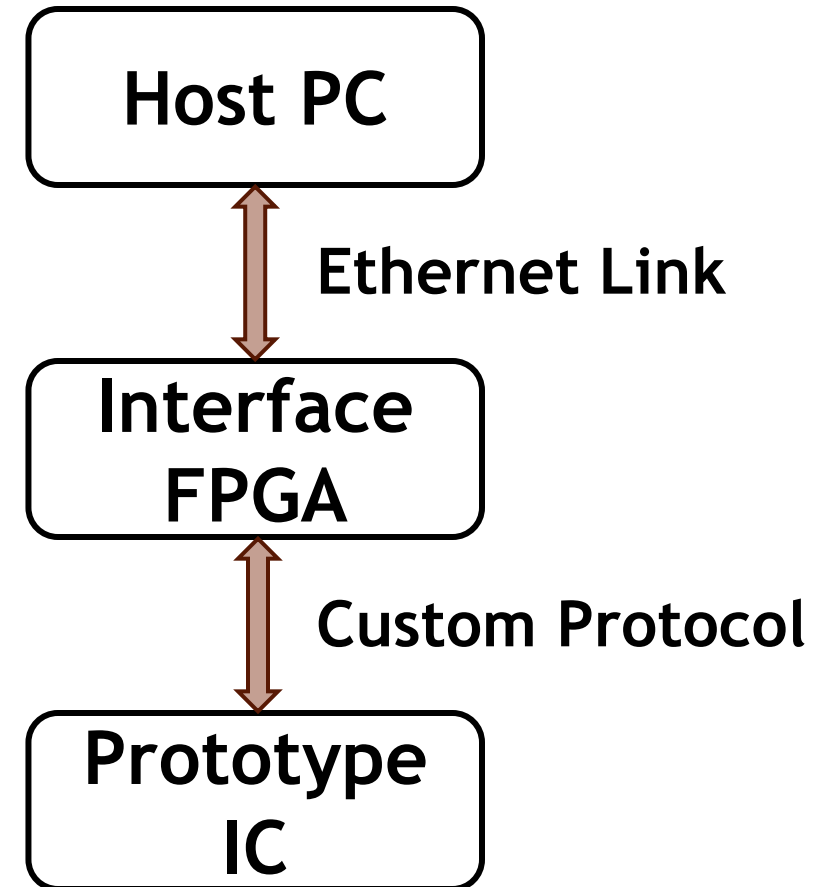
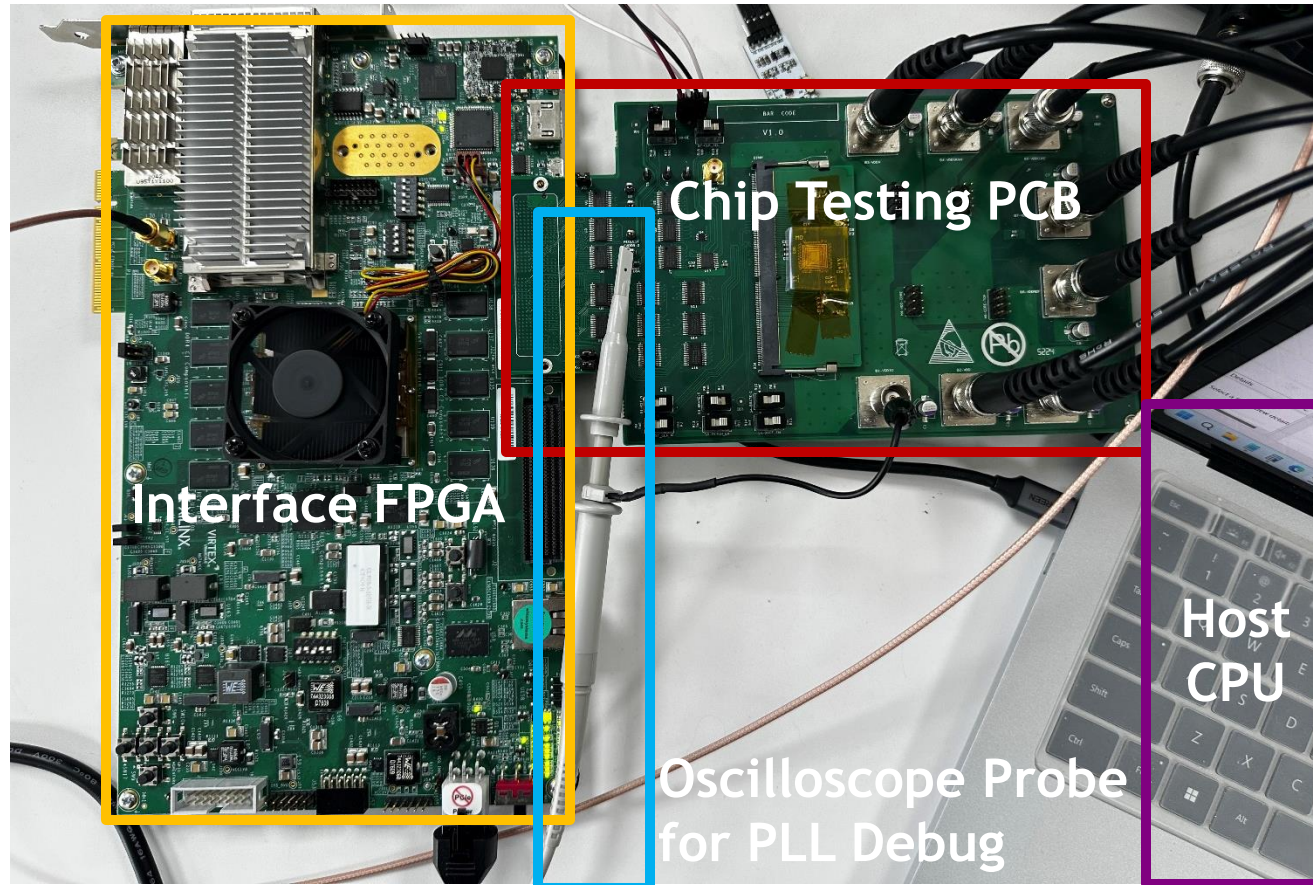
# Area Breakdown



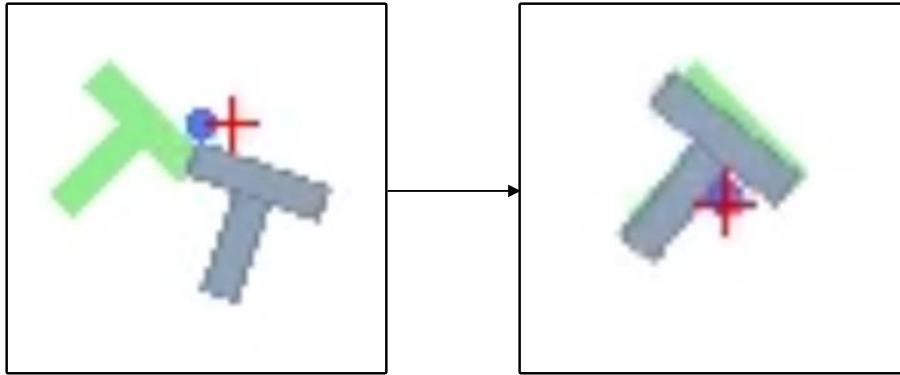
Minimal area overheads from speculation and disturbance injection



# System Demonstration



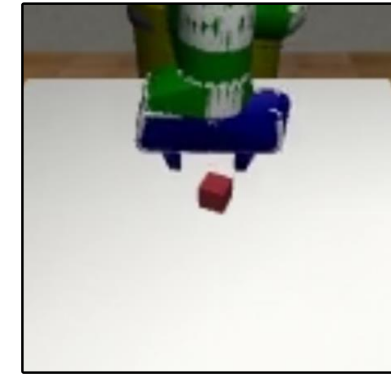
# Evaluated Benchmarks & Configurations



Push-T <sup>1</sup>



Kitchen <sup>2</sup>



Lift <sup>3</sup>

Sequence length = 10  
#Transformer layers = 5  
Embedded dimension = 256  
Speculation depth = 8  
Original denoising steps = 100

**Fully on-chip**

12

8

768

8

100

10

8

256

8

100

<sup>1</sup> [Florence, CoRL'21], <sup>2</sup> [Mandlekar, CoRL'21], <sup>3</sup> [Gupta, CoRL'19]

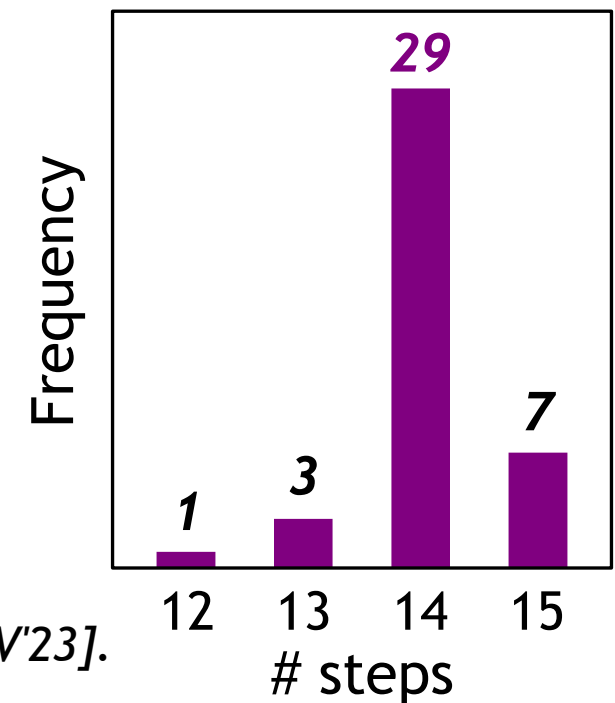


# Inference Performance

- Achieve ~58x speed up with negligible performance loss
- Demonstrate a high commit rate of long intervals

| Task    | Orin AGX  |         |       | This Work*<br>(@1.25GHz) |       |
|---------|-----------|---------|-------|--------------------------|-------|
|         | Precision | Latency | Score | Latency                  | Score |
| Push-T  | INT8      | 391 ms  | 0.75  | 6.8 <sup>**</sup> ms     | 0.74  |
| Lift    | FP32      | 7451 ms | 1.0   | 18.2 <sup>***</sup> ms   | 1.0   |
| Kitchen | FP32      | 8289 ms | 0.56  | 71.2 <sup>***</sup> ms   | 0.52  |

Step distribution in Push-T



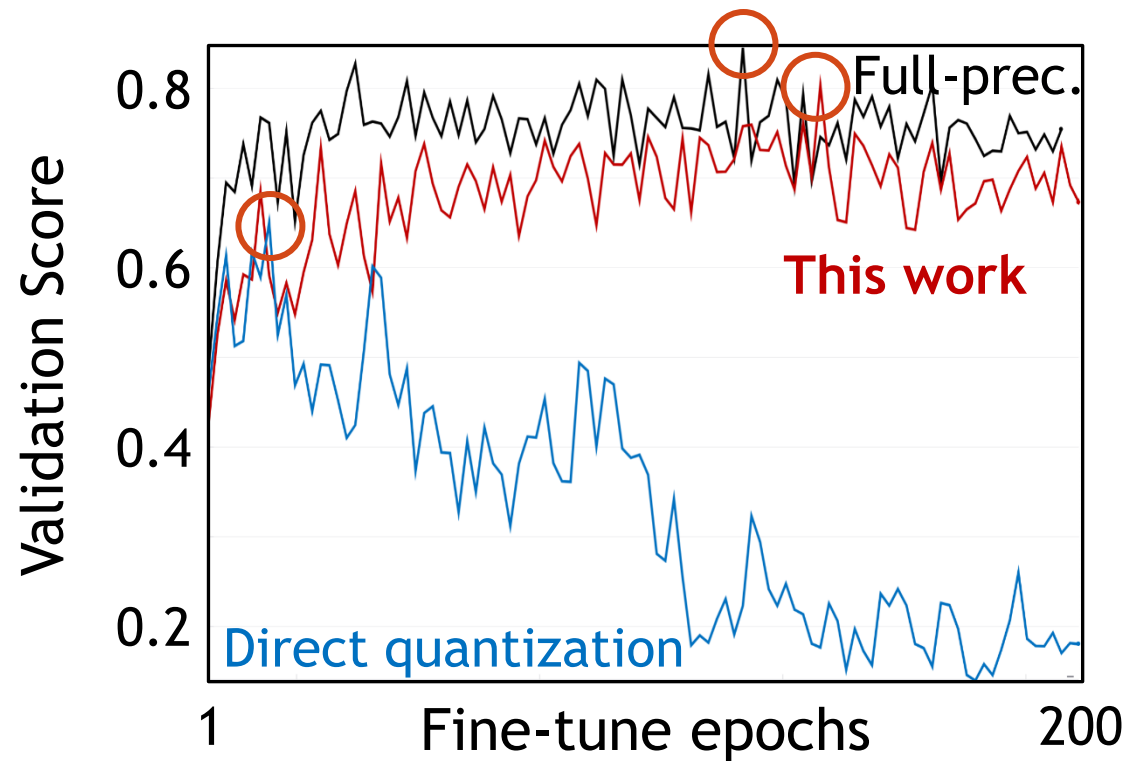
\* Weights are quantized to FP4 using a similar framework to Q-Diffusion [Li, ICCV'23].

\*\* Fully on-chip. \*\*\* Including latencies for off-chip memory access.

# Fine-tune Demonstration

- Efficient low-bit fine-tune w/ better stability

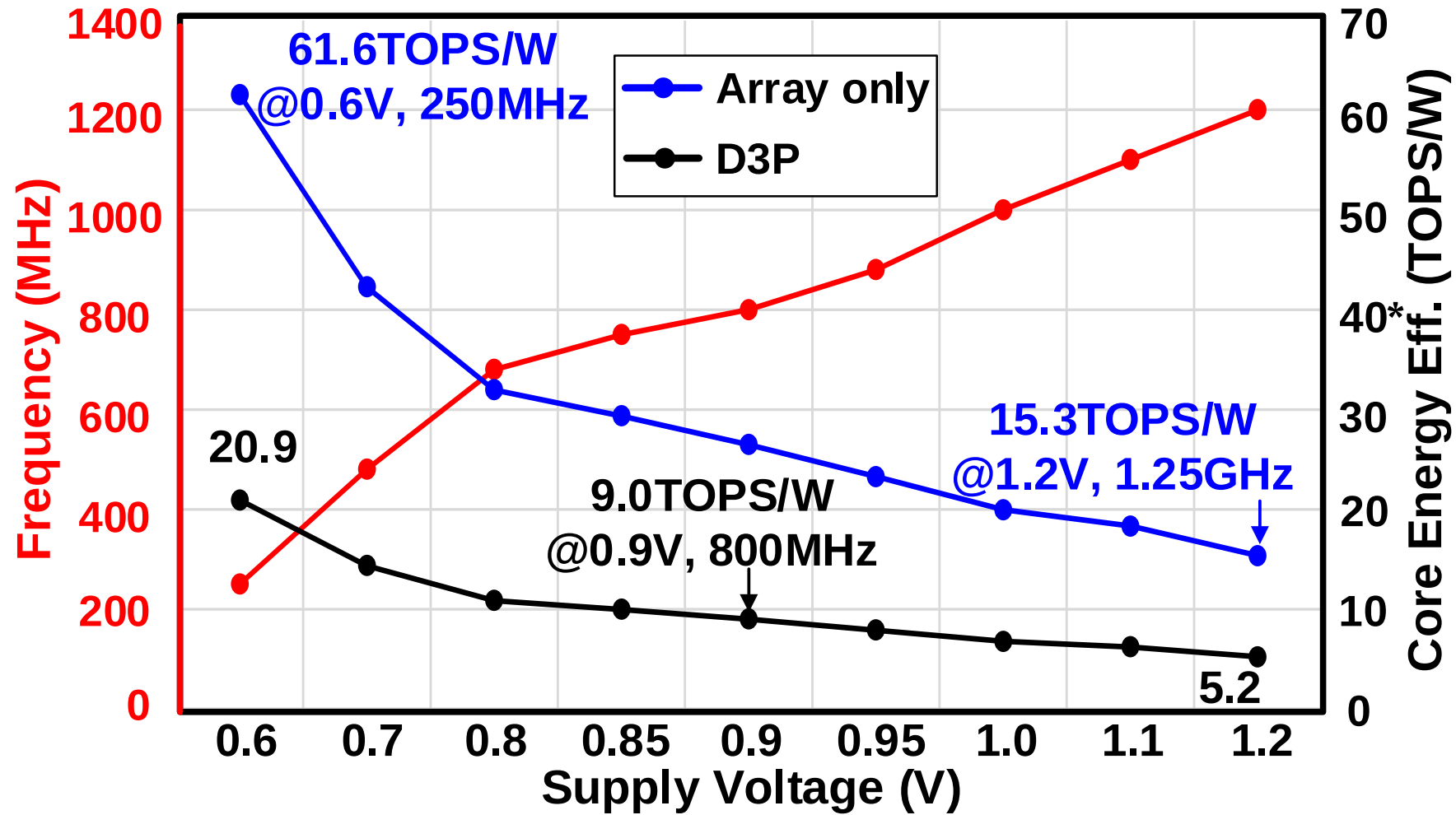
Fine-tune performance curve on Push-T



| Task         | Push-T        | Kitchen          | Lift |
|--------------|---------------|------------------|------|
| New Config.  | Light & Color | Camera Extrinsic |      |
| No fine-tune | 0.15          | 0.0              | 0.0  |
| Full-prec.   | 0.76          | 0.56             | 1.0  |
| Direct INT8* | 0.52          | 0.43             | 1.0  |
| This work *  | 0.74          | 0.55             | 1.0  |

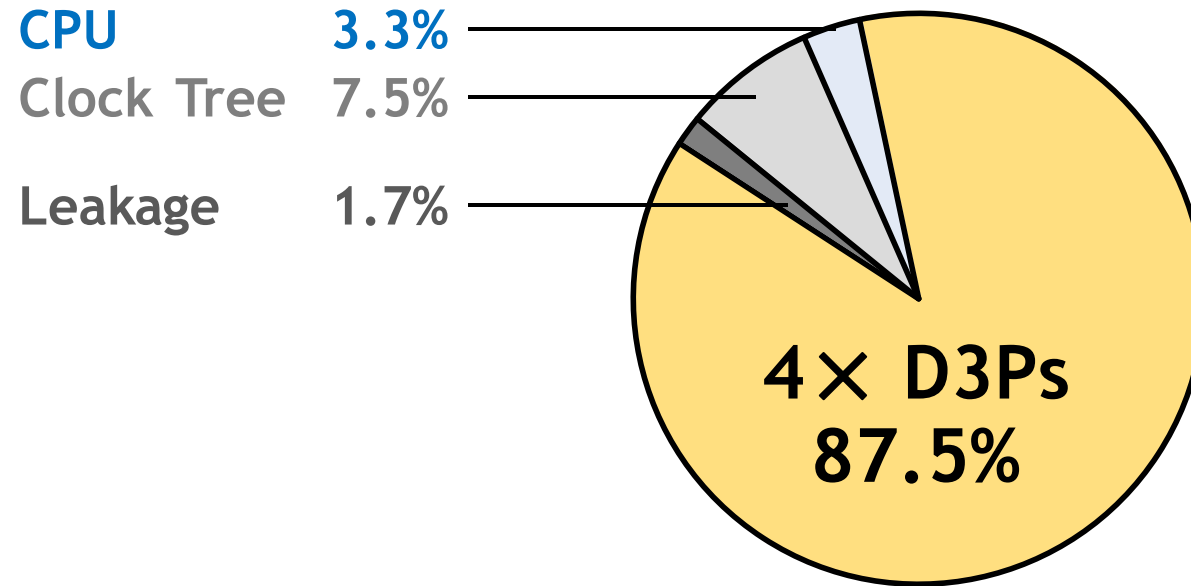
\* Forward & backward: INT8, optimizer: FP16

# Voltage Scaling



\* Feedforward layers in Push-T task, w/ 80% input activation sparsity and 50% weight sparsity, and both in INT8

# System Energy Breakdown



Modest overheads  
from weights loading  
thanks to speculation

| D3P Energy Breakdown [fJ/OP]<br>(1024×256 linear layer w/ 8b/8b weight/activation, @0.9V, 800MHz) |      |               |      |
|---------------------------------------------------------------------------------------------------|------|---------------|------|
| Activation Buffer                                                                                 | 15.1 | Weight Buffer | 11.4 |
| PTE + MAC Array                                                                                   | 37.8 | PPU           | 13.1 |
| Accumulation Buffer                                                                               | 27.8 | Other         | 5.9  |

# Comparison with State-of-the-art

Support training

|                                             | Qin, VLSI'24       | Yoo, ESSERC'24         | Cao, VLSI'24     | Yang, CICC'23 | <b>This Work</b>             |
|---------------------------------------------|--------------------|------------------------|------------------|---------------|------------------------------|
| Technology                                  | 22nm               | 28nm                   | 65nm             | 28nm          | <b>28nm</b>                  |
| Application                                 | Image generation   | Image/Video generation | Robot locomotion | DNN training  | <b>Robot manipulation</b>    |
| Area [mm <sup>2</sup> ]                     | 3.70               | 21.9                   | 3.34             | 3.24          | <b>8.41</b>                  |
| Voltage [V]                                 | 0.6-1.0            | 0.62-1.2               | 0.7-1.2          | 0.52-1.14     | <b>0.6-1.2</b>               |
| Frequency [MHz]                             | 120-540            | 25-400                 | 1                | 50-425        | <b>250-1250</b>              |
| Power [mW]                                  | 60-279             | Not Reported           | 0.432            | 31-1519       | <b>114-2260</b>              |
| Precision                                   | FP16               | HYP8/INT8-13           | -                | Mixed BFP 0-8 | <b>FP4,INT4,INT8</b>         |
| Area Efficiency [T(FL)OPS/mm <sup>2</sup> ] | 3.25               | 0.125                  | -                | 0.27 (8b)     | <b>0.24-1.20<sup>1</sup></b> |
| Infer. Energy Eff. [T(FL)OPS/W]             | 52.01 <sup>4</sup> | 4.96                   | 3.2-6.5 (8b)     | -             | <b>17.6<sup>1,2</sup></b>    |
| Train. Energy Eff. [T(FL)OPS/W]             | -                  | -                      | -                | 3.29 (8b)     | <b>10.9<sup>2,3</sup></b>    |

<sup>1</sup> Measured on Push-T task. <sup>2</sup> Measured at 0.6V, 260MHz. <sup>3</sup> W/ 80% input activation sparsity and 50% weight sparsity, no inputs similarity. <sup>4</sup> Sparse equivalent.

# Outline

- Introduction
- Challenges of diffusion-transformer-based action generation (DiTAG)
- Edge SoC for DiTAG in robot manipulation
  - Overall architecture
  - Dynamic-time-step speculation-commitment dispatcher (SCD)
  - Differential-input dynamic-bit-width diffusion processor (D3P)
  - On-chip randomness-reduction avoidance
- Measurements and demonstrations
- Conclusion

# Conclusion

- Robot manipulation requires low-latency action generation and on-site fine-tuning for better interacting with and adapting to the physical world.
- DiTAG demonstrates better task performance while suffering from deep iterations and quantization-induced randomness reduction.
- Microarchitecture enables memory-access reduction by speculative parallel execution, and avoids randomness reduction with disturbance injection.
- Prototype SoC in 28nm shows  $>30\times$  action-generation throughput and low-cost on-device fine-tune.

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**Thanks for your attention!**